Concern Resolution Process for In-house Manufacturing

QC Story Project

To eliminate concern of front stab bar bolt/nut not fit/slip in all model.

QC STORY PROJECT

PROJECT TITLE :-

To eliminate concern of front stab bar bolt not fit in all model.

Mahindra and Mahindra Ltd, Nasik Plant, Scorpio TCF.

	Plan	Actual
Start Date	01.05.19	20.05.19
Completion Date	30.8.19	25.9.19

Current Status:

Closed

Phase II – QC Story

- Team Information
- Step1: Select the Theme
- Step 2: Justify the choice
- Step 3: Understand the current situation
- Step 4: Select Targets
- Step 5: Analysis (Level 1 Analysis)
- Step 6: Implement corrective measures
- Step 7: Confirm the Effects
- Step 8: Standardize
- Step 9: Summarize & Plan future actions
- Guidelines to user

1. Select the Theme

Brief description of Problem :-To eliminate concern of front stab bar bolt not concern in all model .

What :- Front stab bar bolt/nut not fit/slip in all model.

Where :- Front stab bar bolt fitment stage.

When :- From March -2019

Why **:-** V1 concern and Major contributor in FRC loss, Rework is difficult.

Who :- Line operator.

How :- 3-4 defect/day.

Front stab bar bolt not fit on Chassis at axle stage

2. Justify the choice (reasons)

PROMISE 2022
AUTOMOTIVE SECTOR

We Will Grow

Revenue 75% | Profit 100%

By

'Enhancing lifestyle' in personal segment
'Enriching livelihood' in commercial segment

Being #1 in UV, SCV, EV and #3 in HCV
Doubling volume from overseas markets

Leading in next generation mobility solutions

#InspiringPeopleToRise

Crusade 2.0

Departmental Objective :-
BIQ – DPR Improvement
- FRC / FBO Improvement

Cross Functional Team

No.	Description	Name	Department	Mobile Number
1.	**Team Leader**	**A B Chaudhari.**	**S/X TCF PRODUCTION**	**7588616388**
2.	Mentor	Taj Shaikh	S/X TCF PRODUCTION	8378991344
3.	Member 1	Shailesh Kumar	S/X TCF SQA	8888873405
4.	Member 3	Santosh Jadhav	S/X TCF MQA	9423309400
5.	Member 3	Jalindar Jadhav	S/X TCF PRODUCTION	9421507914
6.	Member 4	Sayali More	S/X TCF Process Engg.	7262090356

2. Justify the choice (reasons)

Problem Criticality (Severity)

- **Importance & Severity of Concern :-**

- **Front stab bar bolt/nut not fit is V1 concern.**
- **Major contributor in FRC loss concern.**
- **Skill operator required for rework.**
- **Rework activity is critical.**
- **Repair cost is very higher 229238.7 Rs. per year.**

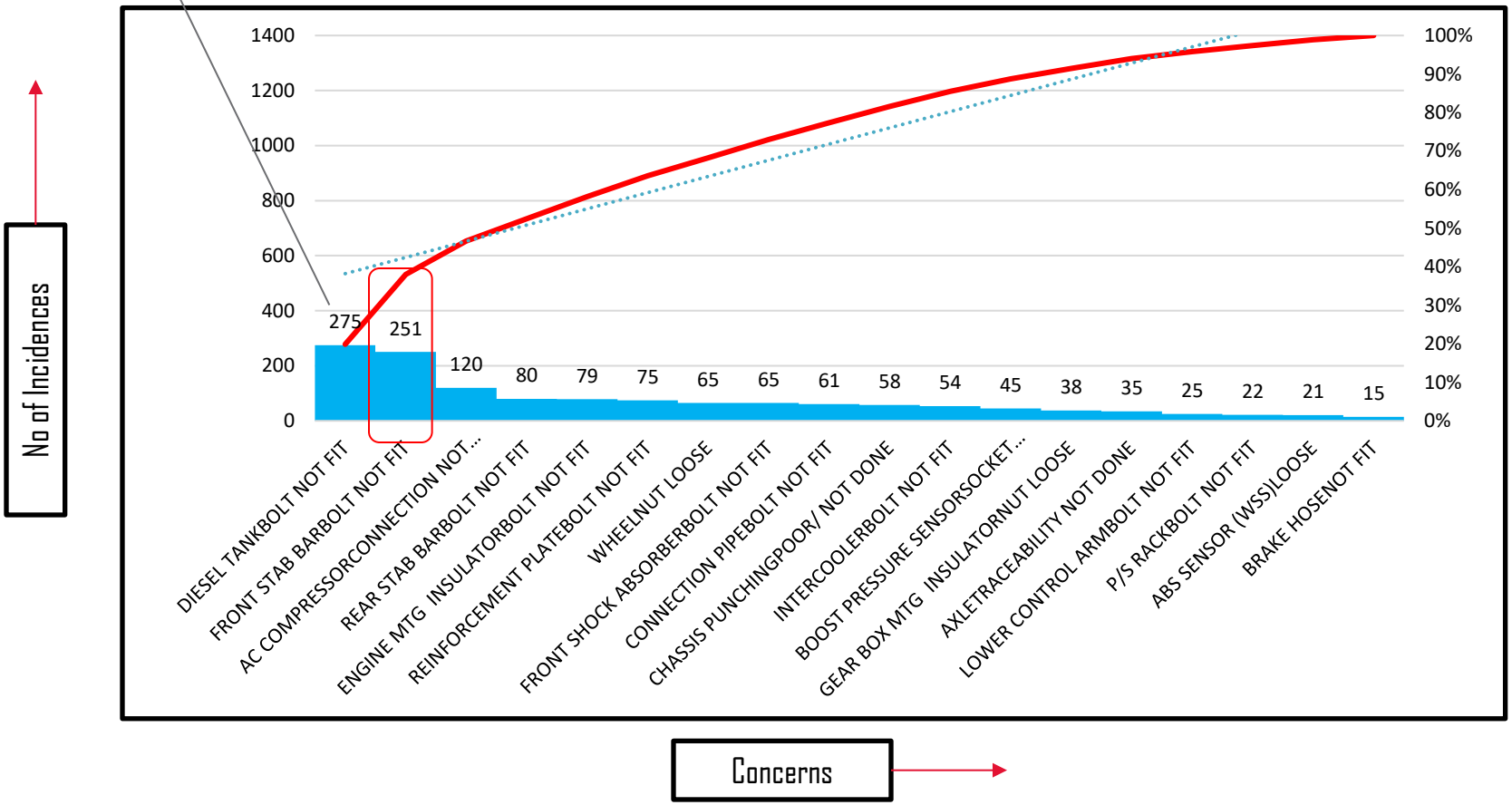
Rework Cost Calculations :-

Repair /Restoration Cost estimation		
Description	Units	
Rework time (Approx)	Min	45
Avg Concerns/Day	Nos	4.0
Total Required Rework time per day	Min	180
Effective working time	Min	465
Operator required /day	nos	0.39
Approx Operator cost/ day	Rs	1974
Operator cost /concern	Rs	764.13
Avg no. of working days	days	25
Operator cost /month	Rs	19103.2
Operator Cost/year	**Rs**	**229238.7**

2. Justify the choice (reasons)

Defect assigned to other officer

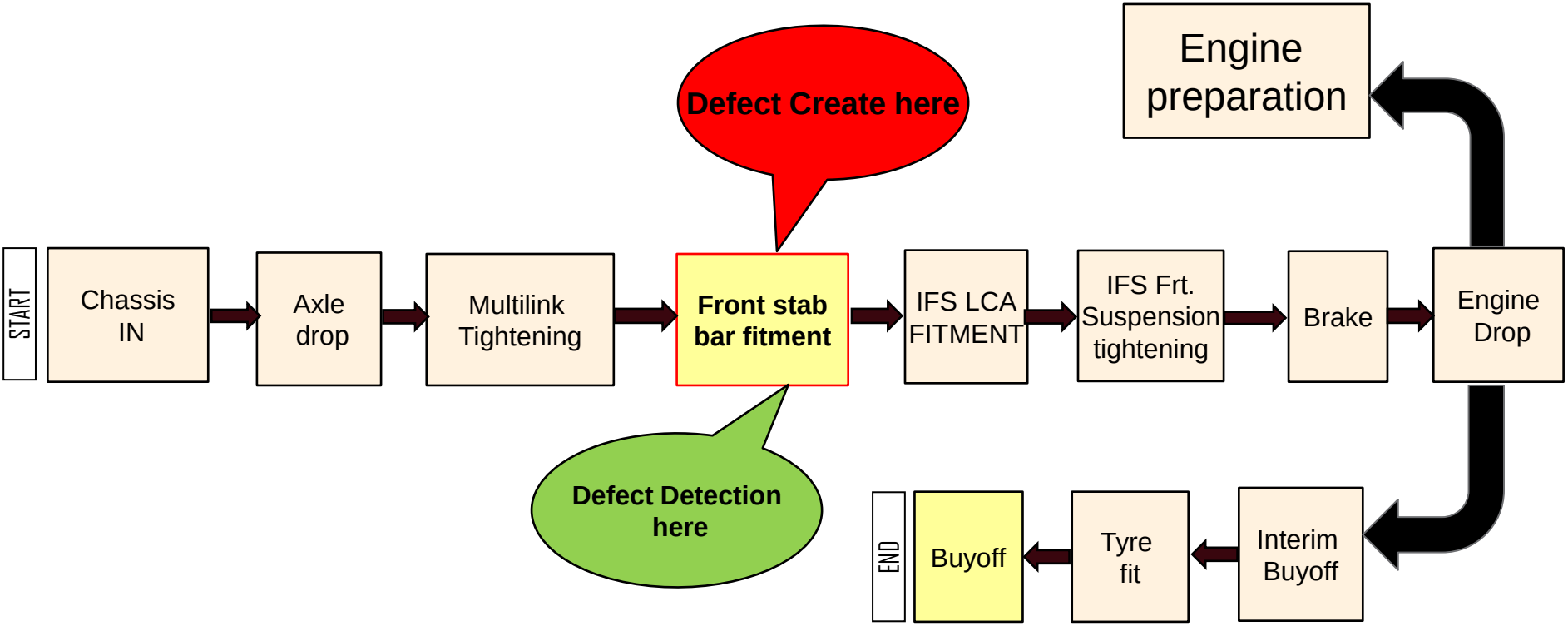
Chassis line defects March to May'19



Inference – After Pareto analysis it is observed that front stab bar bolt/nut not fit/ slip is second top concerns in QDMS Buy off

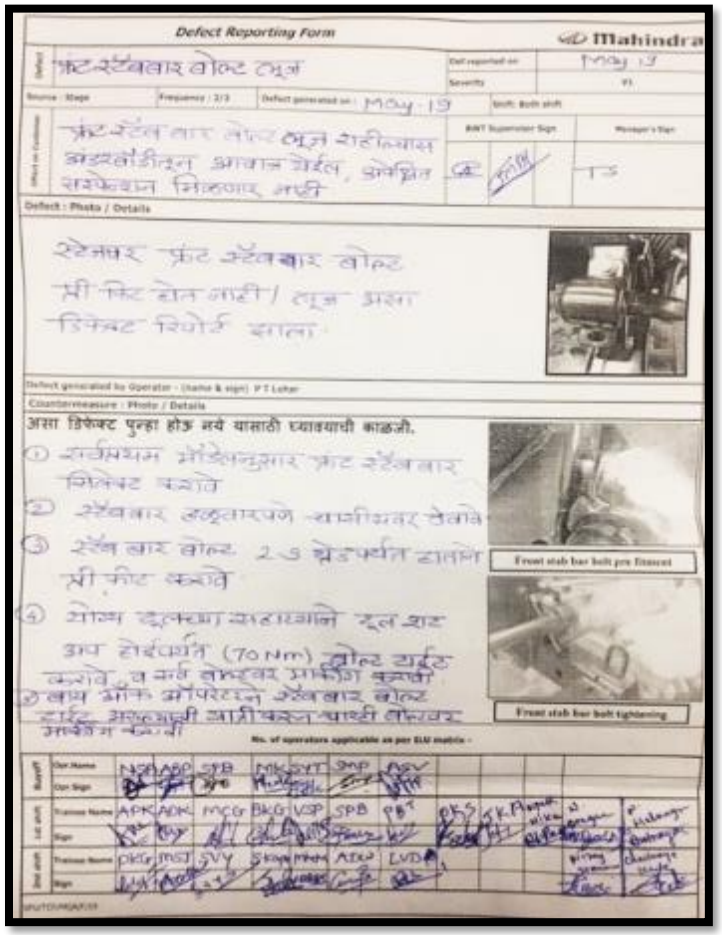
3. Understand the current situation

Observation of process flow – of Chassis line



3.a Immediate Containment Action

- DRF given to operators for Concern detection at stage & buyoff.



Defect communication done through DRF to all stage associate. & Buy off associate.

3. Understand the current situation

Observation of front stab bar fitment

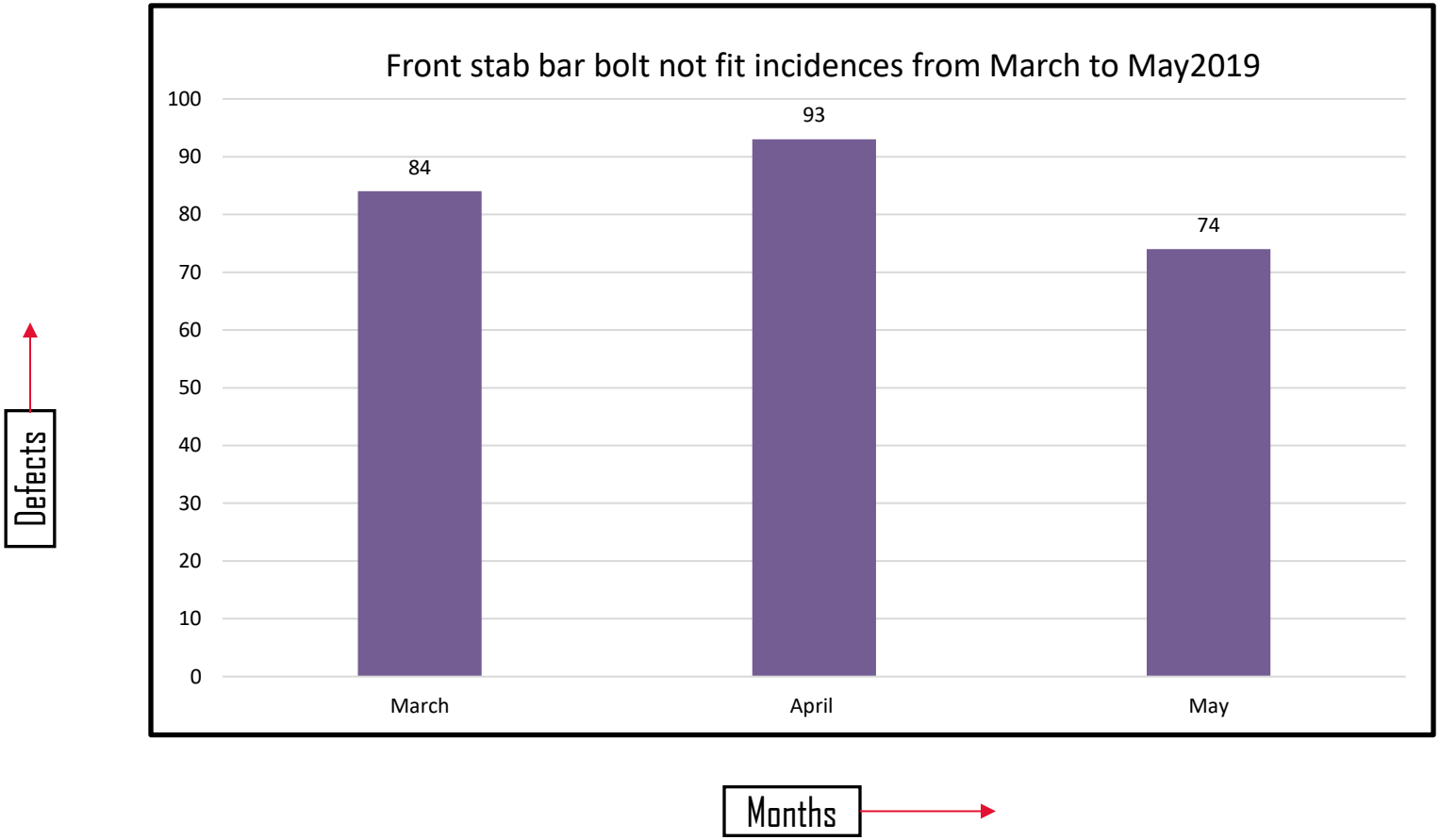
Pick up Front stab bar from storage

Positioning Front stab bar on chassis

Front stab bar bolt pre fitment

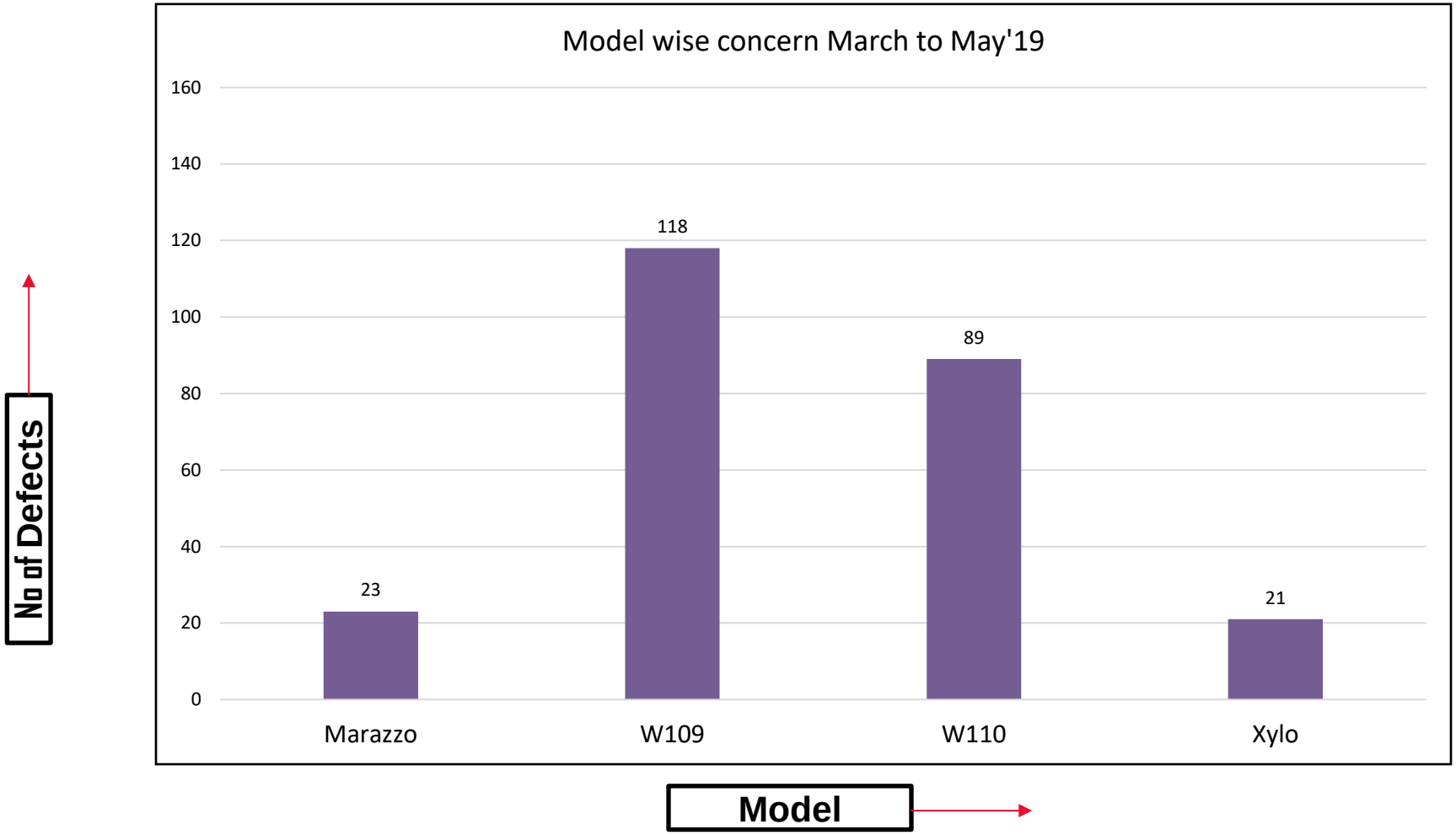
Front stab bar bolt tightening

3. Understand the current situation



Inference- Average 3/4 vehicles per shift reported for Front stab bar bolt not fit defect.

3. Understand the current situation



Inference- Scorpio model is most affected model for this issue.

3. Understand the current situation

4M Analysis (Part 1)
(Conformity Analysis - 'L0')

PU/ Area : S/X TCF- Chassis
Cell / BWT / Line: 7

Requested By - MQA | Date - 03.06.2019 | Source - Stage | 4 M Analysis Part 1 Sr.No :

Problem / Concern Statement : Front stabilizer bar bolt not fitted in all model

FMEA Severity /Gravity : ☐ V1+ ☑ V1 ☐ V2 ☐ V3 | No of Incidences on the day (Including Yard check) : 5 | VIN / Aggre. No.- | Model : W110/ W109

Date of Incidence/ Assy : 03.06.2019 | Shift / Time : Both | Whether SC/CC/Z parameter affected ? ☑ YES ☐ NO ☐ Z ☐ CC ☑ SC

Immediate containment action for Customer Protection :
WIP Chassis checked, Defective chassis taken out.
WIP Stab bar checked for front stab bar 'C' clamp posotion

Does a related 4M Analysis - Part 1 or QC story already exist ? ☐ YES ☑ NO | If Yes | 4M Analysis Part 1 Sr. No. | QC Story Sr. No

Failure mode or cause/s identified -

Problem Assigned- Production | BWT . :

Does a control already exist ? ☐ YES ☐ NO

Stage / Post Analysis -

Man(Human Action)

Operator Name : A P Kalpande/ T D Karambelkar
Is operator trained ? ☑ YES ☐ NO
Operator level (I.L.U) : Black U/ Black L
OR Dexterity level(1…4) : L3
Is operator under training ? ☐ YES ☑ NO
Is operator temporary for Stage ? ☐ YES ☑ NO
Bad rework ? ☐ YES ☑ NO

Method(process)

Main steps are respected ? ☑ YES ☐ NO
Is there a related key point in SOP / SOS? ☑ YES ☐ NO
Are the reasons for key point know n ? ☑ YES ☐ NO
Are operating reqmnts respected ? ☑ YES ☐ NO
Is the defect part of the Buy off / Checkman list ? ☑ YES ☐ NO
Was it detected by Inspector/ Buoff Opr ? ☐ YES ☐ NO

Other

Is there a Control Plan ? (Record in Analysis table) ☑ YES ☐ NO
Is the packaging standard defined for the part ? ☑ YES ☐ NO
Is the packaging standard respected ? ☑ YES ☐ NO
Is there any alert mechanism for failure ? ☐ YES ☑ NO
Buy off -A B Patel/ R B Mahajan

Machine

Is the specified /Tool / Fixture facility used ? ☑ YES ☐ NO
Last calibration dates
Facility /Tool / Fixture in good condition ? ☑ YES ☐ NO
Is it a substitute facility ? ☐ YES ☑ NO
Are machine parameters know n ? (Record in Analysis table) ☑ YES ☐ NO
Parameters are respected ? ☑ YES ☐ NO

Product Failure

In only one shift ? ☐ YES ☑ NO
In only one line ? ☑ YES ☐ NO
The problem follow s the vehicle ? ☐ YES ☑ NO
The problem follow s the part ? ☑ YES ☐ NO
Is it a repeat problem ? ☑ YES ☐ NO
Is the suspected parameter measurable ? (Record in Analysis table) ☑ YES ☐ NO

Material/ Consumable

Is the specified product used ? ☑ YES ☐ NO
Did it evolve / Change ? ☐ YES ☑ NO
Is the storage SOP respected ? ☑ YES ☐ NO
Is the shelf life / Expiry date respected ? ☑ YES ☐ NO
Is the part as per FIFO? ☑ YES ☐ NO

Note : 1. If both fields (Yes/NO) are kept unmarked, it means that mentioned factor is not applicable for this concern)

Analysis Conclusion (Attach Sketch/ Photograph if necessary) -

Front stab bar bolt not fit

History - These vehicles are reported at stage for front stab bar bolt not pre fit in both shifts.
Process flow - Stage associate check build sheet to confirm model, pick the front stab bar from sequencing trolley confirming stab bar is as per model, install front stab bar on chassis, pre fit stab bar bolt on chassis bracket, IFS stage associate tight the stab bar bolt by pulse tool till tool shut off.

Detection - Check point of 'stab bar bolt not loose & stab bar is as per model' at buy off

Traveler card - NIL.
Packaging - NIL.
Defect history - NIL
Failed part : Chassis/ Front stab bar

Analysis : - 1. After analysis we found trained associate deployed, he is well known about sos, key point and main step,
2. During stab bar installation on chassis stab bar bolt not getting pre fit on chassis bracket, Chassis stab bar mounting weld nut found rusty.
3. Stab bar 'C' clamp CD not getting match with Stab bar bolt mounting bracket,
4. Stab bar bolt remains half torque & some times bolt get slip

Sno	Product / Process / Machine / Parameter	Spec.	Actual	Remark
1	Front stab bar bolt tightening	50 +/- 5 Nm	51 Nm	OK

Is 4M Analysis Part 2 required ? ☑ YES ☐ NO

Is the substitute transport mode used ? ☐ YES ☑ NO **Is the packaging defined ?** ☑ YES ☐ NO

Actions -

1.Containment Action : -

	Resp	Date	Status
1. DRF given to all stage associate	AC/MP	04.06.2019	Closed

2. Permanent Action : -

	Resp	Date	Status
1. PART CORRECTION TO BE DONE.	Shailesh Kumar	15.07.19	

Analysed by Name/s - Ajit Chaudhari
Date - 04.06.19

Choice : ☑ WSS ☐ QC Story

L0 for front stab bar bolt/nut not fit/slip in all model

Inference from L0
Problem follows part & vehicle
Problem is a repeat problem
L1 analysis required.

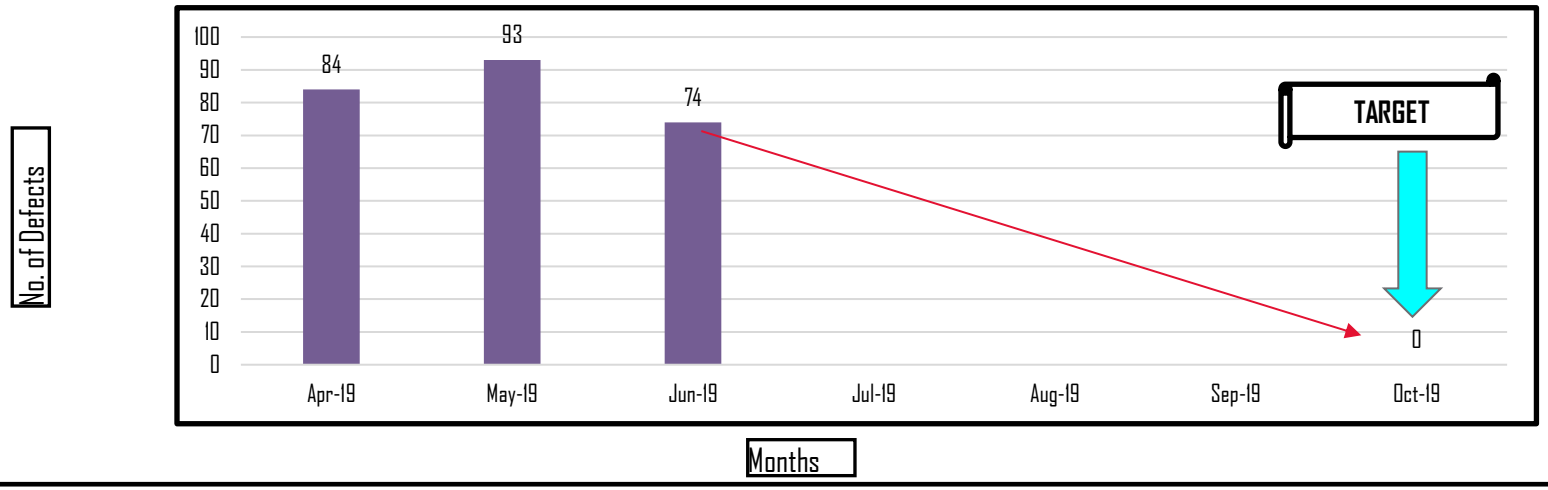
4M analysis Good and Not so good observations	
MAN	Operator dext. Level L3/L4
Method	Front stab bar bolt pre fitment hard
Others	Control plan adhered on stage
Machine	Tools used are calibrated
Product failure	Problem follows part
Material	Material stored as per requirement.

4. Select Target

Identify the target. A target is a number indicating the level of improvement to be achieved.

Present Level:	Daily 3-4 vehicles reported for front stab bar bolt not fit concerns.
Target:	**To eliminate concern of front stab bar bolt/nut not fit/slip in all model up to OCT 2019.**

Target setting for front stab bar bolt not fit concern



Step 5. Analysis

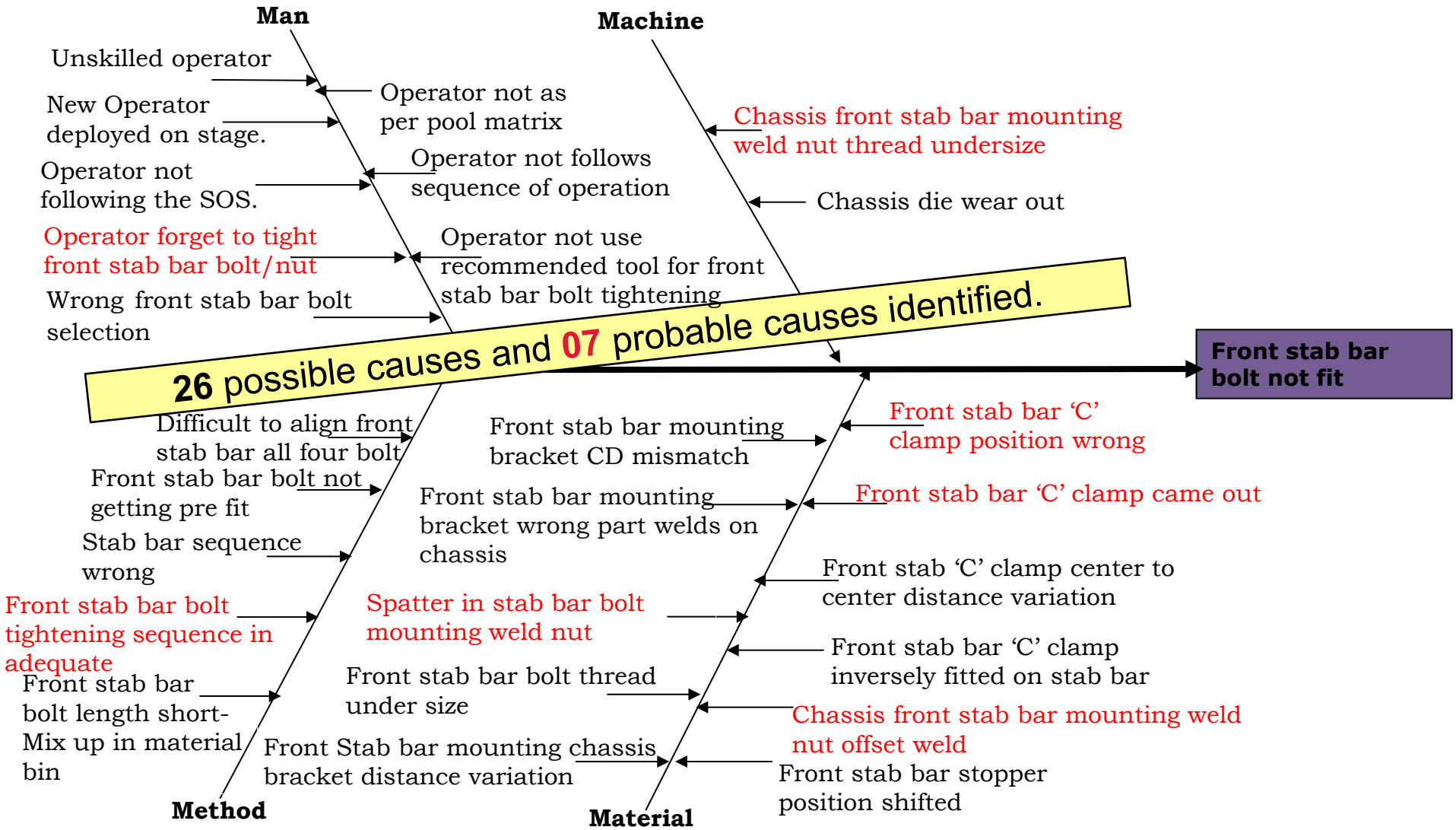
List of Possible causes after brainstorming among team members

Sr. no	Possible Causes
1	Unskilled operator
2	New Operator deployed on stage.
3	Operator not following the SOS.
4	Operator forget to tight front stab bar bolt/nut
5	Operator not as per pool matrix
6	wrong position of stab bar fitment
7	Operator not use recommended tool for front stab bar bolt tightening
8	Front stab bar mounting bracket CD mismatch
9	Front stab bar mounting plate welds in inverted condition on chassis
10	Front stab bar bolt tightening tool inoperative.
11	Wrong front stab bar bolt selection.
12	Spatter in stab bar bolt mounting weld nut
13	Front stab bar bolt thread under size

Sr. no	Possible Causes
14	Front stab bar mounting chassis bracket damage
15	Front stab bar 'C' clamp position wrong
16	Front stab bar bolt pre fitment hard on chassis bracket
17	Front stab 'C' clamp center to center distance variation
18	Front stab bar 'C' clamp inversely fitted on stab bar
19	Chassis front stab bar mounting weld nut offset weld
20	Chassis front stab bar mounting weld nut thread undersize
21	Difficult to align front stab bar all four bolt
22	Front stab bar 'C' clamp came out
23	Front stab bar bolt length short
24	Stab bar sequence wrong
25	Front Stab bar mounting chassis bracket distance variation
26	Chassis die wear out

5. Analysis

5.1 Identify possible causes: Fish Bone diagram.



5. Analysis

5.1 List of possible causes

Sr. no	Possible Causes	Ajit	Taj Shaikh	Shailesh Kumar	Mandar padwal	D Jadhav	J Jadhav	Santosh	Total
1	Unskilled operator	1	1	3	3	3	3	3	17
2	New Operator deployed on stage.	1	1	1	1	1	1	1	7
3	Operator not following the SOS.	3	1	3	3	3	9	3	25
4	Operator forget to tight front stab bar bolt/nut	3	9	9	3	3	9	3	39
5	Operator not as per pool matrix	1	3	9	1	1	3	1	19
6	wrong position of stab bar fitment	1	1	1	3	1	1	1	9
7	Operator not use recommended tool for front stab bar bolt tightening	1	1	3	3	3	3	3	17
8	Front stab bar mounting bracket CD mismatch	3	1	3	3	3	9	3	25
9	Front stab bar mounting plate welds in inverted condition on chassis	1	1	1	1	1	1	1	7
10	Front stab bar bolt tightening tool inoperative.	1	1	1	1	1	1	1	7
11	Wrong front stab bar bolt selection.	1	1	3	3	3	3	3	17
12	Spatter in stab bar bolt mounting weld nut	9	9	9	3	3	9	3	45
13	Front stab bar bolt thread under size	3	1	3	1	3	1	3	15

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 9- Significant impact.
Threshold Value for selecting cause for hypothesis is 39.

5. Analysis

5.1 List of possible causes

Sr. no	Possible Causes	Ajit	Taj Shaikh	Shailesh Kumar	Mandar padwal	D Jadhav	J Jadhav	Santosh	Total
14	Front stab bar mounting chassis bracket damage	1	1	1	3	1	1	3	11
15	Front stab bar 'C' clamp position wrong	3	9	3	9	3	9	3	39
16	Front stab bar bolt pre fitment hard on chassis bracket	3	1	3	3	3	9	3	25
17	Front stab 'C' clamp center to center distance variation	1	1	3	3	3	3	3	17
18	Front stab bar 'C' clamp inversely fitted on stab bar	3	1	3	3	3	9	3	25
19	Chassis front stab bar mounting weld nut offset weld	3	9	9	3	3	9	3	39
20	Chassis front stab bar mounting weld nut thread undersize	3	3	9	3	9	3	9	39
21	Difficult to align front stab bar all four bolt	9	9	9	3	3	9	3	45
22	Front stab bar 'C' clamp came out	1	1	3	3	3	3	3	17
23	Front stab bar bolt length short	1	1	3	3	1	3	3	15
24	Stab bar bolt tightening sequence inadequate	9	9	9	3	3	9	3	45
25	Front Stab bar mounting chassis bracket distance variation	1	1	1	3	1	1	3	11
26	Chassis die wear out	1	9	3	3	1	3	3	23

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 9- Significant impact.
Threshold Value for selecting cause for hypothesis is 39.

Step 5. Analysis

List of causes not considered as probable causes and reasons for not considering

Sr. no	Possible Causes	Ajit	Taj Shaikh	Shailesh Kumar	Mandar padwal	D Jadhav	J Jadhav	Santosh	Total
1	Unskilled operator	1	1	3	3	3	3	3	17
2	New Operator deployed on stage.	1	1	1	1	1	1	1	7
3	Operator not following the SOS.	3	1	3	3	3	9	3	25
5	Operator not as per pool matrix	1	3	9	1	1	3	1	19
6	wrong position of stab bar fitment	1	1	1	3	1	1	1	9
7	Operator not use recommended tool for front stab bar bolt tightening	1	1	3	3	3	3	3	17
8	Front stab bar mounting bracket CD mismatch	3	1	3	3	3	9	3	25
9	Front stab bar mounting plate welds in inverted condition on chassis	1	1	1	1	1	1	1	7
10	Front stab bar bolt tightening tool inoperative.	1	1	1	1	1	1	1	7
11	Wrong front stab bar bolt selection.	1	1	3	3	3	3	3	17
13	Front stab bar bolt thread under size	3	1	3	1	3	1	3	15
14	Front stab bar mounting chassis bracket damage	1	1	1	3	1	1	3	11
16	Front stab bar bolt pre fitment hard on chassis bracket	3	1	3	3	3	9	3	25
17	Front stab 'C' clamp center to center distance variation	1	1	3	3	3	3	3	17
18	Front stab bar 'C' clamp inversely fitted on stab bar	3	1	3	3	3	9	3	25
22	Front stab bar 'C' clamp came out	1	1	3	3	3	3	3	17
23	Front stab bar bolt length short	1	1	3	3	1	3	3	15
25	Front Stab bar mounting chassis bracket distance variation	1	1	1	3	1	1	3	11
26	Chassis die wear out	1	9	3	3	1	3	3	23

5.1 TEST OF HYPOTHESIS

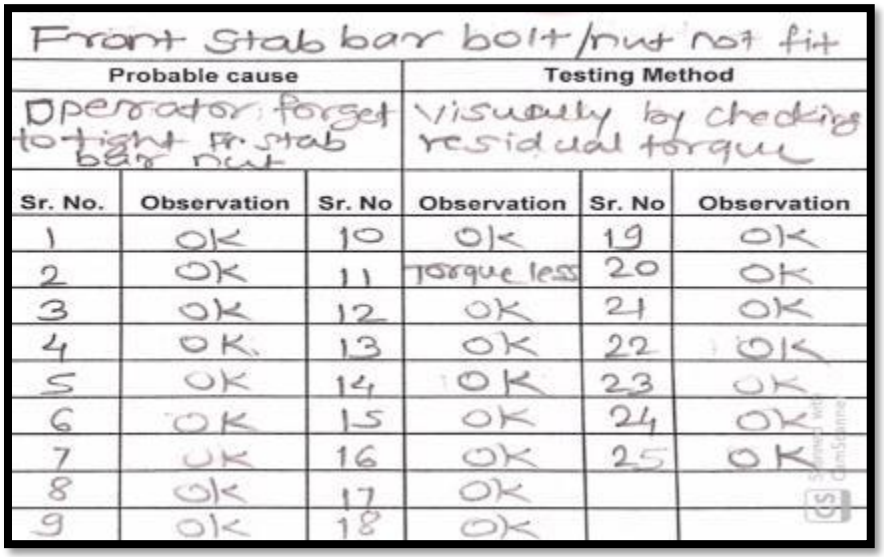
Confirm the probable cause:-

Sr. no	Possible Causes	Ajit	Taj Shaikh	Shailesh Kumar	Mandar padwal	D Jadhav	J Jadhav	Santosh	Total
4	Operator forget to tight front stab bar bolt/nut	3	9	9	3	3	9	3	39
12	Spatter in stab bar bolt mounting weld nut	9	9	9	3	3	9	3	45
15	Front stab bar 'C' clamp position wrong	3	9	3	9	3	9	3	39
19	Chassis front stab bar mounting weld nut offset weld	3	9	9	3	3	9	3	39
20	Chassis front stab bar mounting weld nut thread undersize	3	3	9	3	9	3	9	39
21	Difficult to align front stab bar all four bolt	9	9	9	3	3	9	3	45
24	Stab bar bolt tightening sequence inadequate	9	9	9	3	3	9	3	45

5.3 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
4	Operator forget to tight front stab bar nut	Visually & by checking residual torque	Front stab bar bolt/nut residual torque checked, in **01** vehicle out of **25** vehicles torque found less	**Hypothesis Valid**

Operator forget to tight front stab bar nut



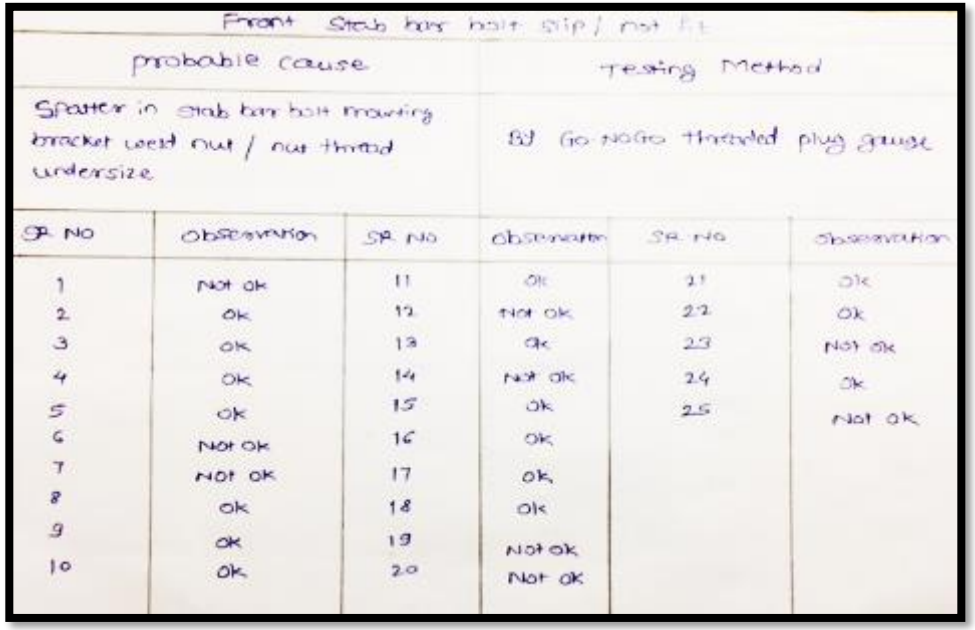
Front Stab bar bolt/nut not fit

Probable cause			Testing Method		
Operator forget to tight fr. stab bar nut			visually by checking residual torque		
Sr. No.	Observation	Sr. No	Observation	Sr. No	Observation
1	OK	10	OK	19	OK
2	OK	11	Torque less	20	OK
3	OK	12	OK	21	OK
4	OK	13	OK	22	OK
5	OK	14	OK	23	OK
6	OK	15	OK	24	OK
7	OK	16	OK	25	OK
8	OK	17	OK		
9	OK	18	OK		

Inference- Front stab bar bolt/nut residual torque checked, in **01** vehicle out of **25** vehicles torque found less, hence this is a Valid cause.

5.3 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
12	Spatter in front stab bar bolt mounting bracket weld nut.	Visually & By Go/No-go threaded plug gauge	Found Spatter in **09** Vehicle during pre fitment out of **25** Vehicle .	**Hypothesis Valid**



Front Stab bar bolt slip / not fit

probable cause	Testing Method
Spatter in stab bar bolt mounting bracket weld nut / nut thread undersize	By Go NoGo threaded plug gauge

SR NO	observation	SR NO	observation	SR NO	observation
1	Not ok	11	Ok	21	Ok
2	Ok	12	Not ok	22	Ok
3	Ok	13	Ok	23	Not ok
4	Ok	14	Not ok	24	Ok
5	Ok	15	Ok	25	Not ok
6	Not ok	16	Ok		
7	Not ok	17	Ok		
8	Ok	18	Ok		
9	Ok	19	Not ok		
10	Ok	20	Not ok		

Inference – Spatter found in 09 Vehicle out of 25 vehicle, hence this is a Valid cause.

5.3 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
15	Front stab bar 'C' clamp position wrong	Front stab bar 'C' clamp CD checking on fixture	Front stab bar 'C' clamp CD check & found shifted in **07** vehicles out of **25** vehicles	**Hypothesis Valid**

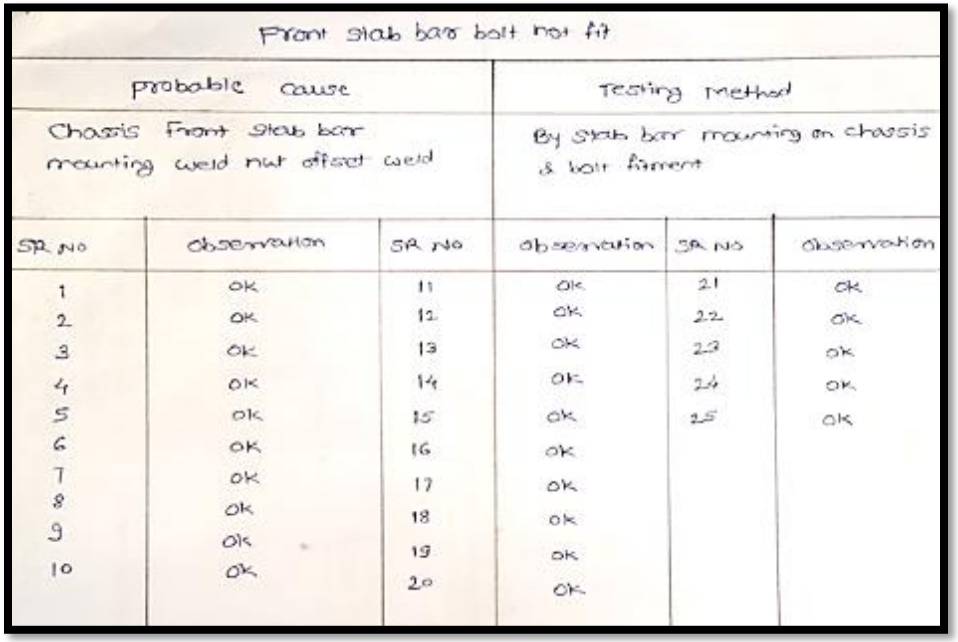
Front Stab bar bolt/nut not fit

Probable cause			Testing Method		
Front stab bar 'C' clamp position wrong / shift			Front stab bar 'C' clamp position/CD checking on Fixture		
Sr. No.	Observation	Sr. No	Observation	Sr. No	Observation
1	OK	10	OK	19	N.OK
2	OK	11	N.OK	20	OK
3	OK	12	N.OK	21	OK
4	OK	13	OK	22	N.OK
5	OK	14	N OK	23	OK
6	N OK	15	OK	24	OK
7	N OK	16	OK	25	OK
8	OK	17	OK		
9	OK	18	OK		

Inference- Front stab bar 'C' clamp position found wrong in 07 Vehicle out of 25 vehicle hence this is a Valid cause.

5.3 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
19	Chassis front stab bar mounting weld nut offset weld	By stab bar mounting on chassis & bolt fitment	Front stab bar mtg. weld nut offset check in 25 vehicles by pre fitment of bolt found all ok out of **25** vehicles	**Hypothesis Invalid**



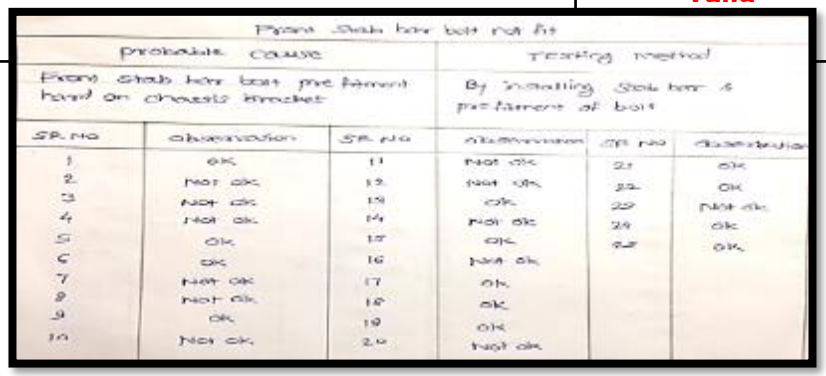
Front stab bar bolt not fit

probable cause			Testing method		
Chassis Front stab bar mounting weld nut offset weld			By stab bar mounting on chassis & bolt fitment		
SR NO	observation	SR NO	observation	SR NO	observation
1	ok	11	ok	21	ok
2	ok	12	ok	22	ok
3	ok	13	ok	23	ok
4	ok	14	ok	24	ok
5	ok	15	ok	25	ok
6	ok	16	ok		
7	ok	17	ok		
8	ok	18	ok		
9	ok	19	ok		
10	ok	20	ok		

Inference- In 25 vehicles, Front stab bar mtg. weld nut offset found ok condition hence this is a invalid cause.

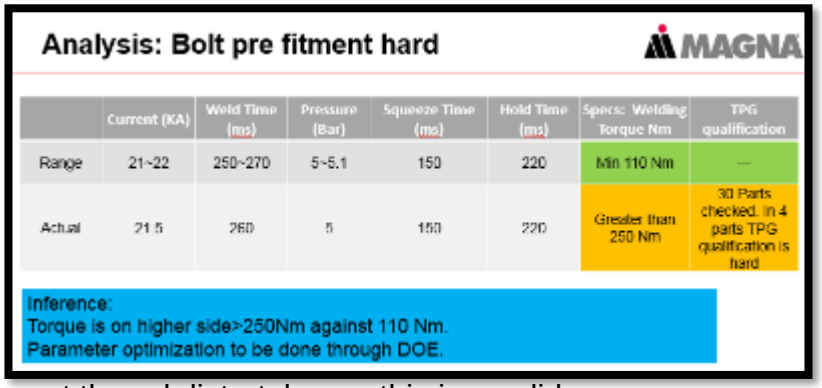
5.3 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
20	Chassis front stab bar mounting weld nut thread distortion /undersize	By actual installing stab bar and pre fitment of bolt	Front stab bar mtg. bolt not getting pre fit in **09** vehicles out of **25** vehicles	**Hypothesis Valid**



Front Stab bar bolt not fit

probable cause			Testing method		
Front stab bar bolt pre fitment hard on chassis bracket			By installing stab bar & pre fitment of bolt		
Sr. no	observation	Sr. no	observation	Sr. no	observation
1	ok	11	Not ok	21	ok
2	Not ok	12	Not ok	22	ok
3	Not ok	13	ok	23	Not ok
4	Not ok	14	Not ok	24	ok
5	ok	15	ok	25	ok
6	ok	16	Not ok		
7	Not ok	17	ok		
8	Not ok	18	ok		
9	ok	19	ok		
10	Not ok	20	Not ok		



Analysis: Bolt pre fitment hard MAGNA

	Current (KA)	Weld Time (ms)	Pressure (Bar)	Squeeze Time (ms)	Hold Time (ms)	Specs: Welding Torque Nm	TPG qualification
Range	21~22	250~270	5~5.1	150	220	Min 110 Nm	—
Actual	21.5	260	5	150	220	Greater than 250 Nm	30 Parts checked. In 4 parts TPG qualification is hard

Inference:
Torque is on higher side>250Nm against 110 Nm.
Parameter optimization to be done through DOE.

Inference- In 09 vehicles, Front stab bar mtg. bolt pre fitment found hard as nut thread distort, hence this is a valid cause.

5.3 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
21	Difficult to align front stab bar all four bolt	Front stab bar bolt pre fitment on Chassis	Front stab bar pre fit on chassis to check Stab bar bolt alignment in **25** vehicles , In all 25 vehicles, Front stab bar bolt aligned on chassis stab bar mounting bracket.	**Hypothesis Invalid**

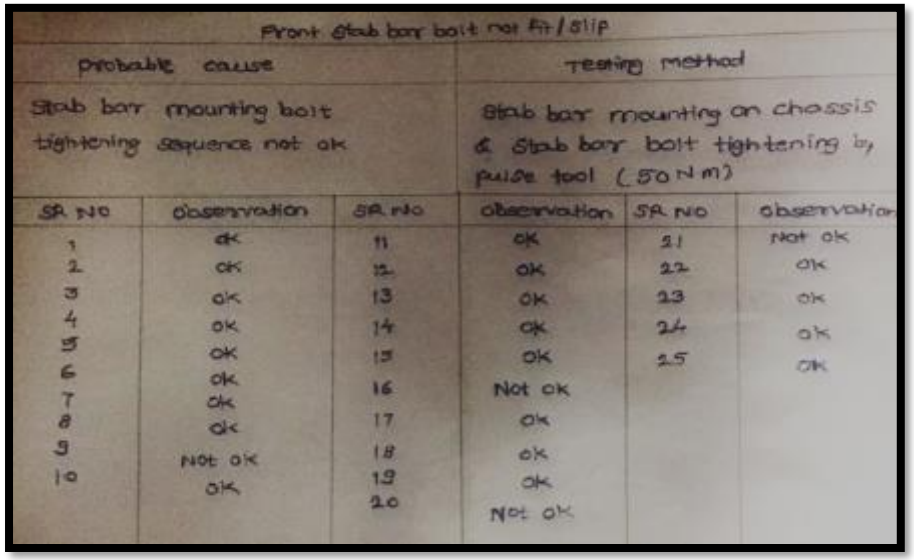
Front Stab bar bolt/nut not fit

Probable cause			Testing Method		
Difficult to alisn Front stab bar all four bolts			Front stab bar & bolt pre fitment on chassis		
Sr. No.	Observation	Sr. No	Observation	Sr. No	Observation
1	OK	10	OK	19	OK
2	OK	11	OK	20	OK
3	OK	12	OK	21	OK
4	OK	13	OK	22	OK
5	OK	14	OK	23	OK
6	OK	15	OK	24	OK
7	OK	16	OK	25	OK
8	OK	17	OK		
9	OK	18	OK		

Inference- In all 25 vehicles, Front stab bar bolt aligned on chassis stab bar mounting bracket, hence it is invalid cause.

5.3 Confirm the probable cause:

No	Probable Cause	Testing method	Observations	Conclusion
24	Front stab bar bolt tightening sequence wrong	Stab bar mounting on Chassis & Stab bar bolt tightening by pulse tool	Front stab bar mtg. bolt tightened on chassis, in **04** vehicles out of **25** vehicles bolt remains half torque	**Hypothesis Valid**



Front Stab bar bolt not fit / slip

probable cause			Testing method		
Stab bar mounting bolt tightening sequence not ok			Stab bar mounting on chassis & Stab bar bolt tightening by pulse tool (50 N m)		
SR NO	observation	SR NO	observation	SR NO	observation
1	ok	11	ok	21	Not ok
2	ok	12	ok	22	ok
3	ok	13	ok	23	ok
4	ok	14	ok	24	ok
5	ok	15	ok	25	ok
6	ok	16	Not ok		
7	ok	17	ok		
8	ok	18	ok		
9	Not ok	19	ok		
10	ok	20	Not ok		

Inference- In 04 vehicles, Front stab bar bolt remains half torque hence it is valid cause.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Conclusion
4	Operator forget to tight front stab bar bolt/nut	Visually & by checking residual torque	Hypothesis valid
12	Spatter in stab bar bolt mounting weld nut	Visually & By Go/No-go threaded plug gauge	Hypothesis valid
15	Front stab bar 'C' clamp position wrong	Front stab bar 'C' clamp CD checking on fixture	Hypothesis valid
19	Chassis front stab bar mounting weld nut offset weld	By stab bar mounting on chassis & bolt fitment	Hypothesis Invalid
20	Chassis front stab bar mounting weld nut thread undersize	By Go/No-go threaded plug gauge	Hypothesis valid
21	Difficult to align front stab bar all four bolt	Front stab bar bolt pre fitment on Chassis	Hypothesis Invalid
24	Stab bar bolt tightening sequence wrong	Stab bar mounting on Chassis & Stab bar bolt tightening by pulse tool	Hypothesis valid

5.3 Identify the root causes: (Use Why – Why)

WHY - WHY Analysis

Sr No	Valid Probable Cause	WHY	WHY	WHY	WHY	ACTION
4	Operator forget to tight front stab bar bolt/nut	Associate only pre fit the all four nut	Associate tight one nut twice & forget to tight last nut	There is no mechanism to confirm all four nut torque	----	Provide the facility
12	Spatter in stab bar bolt mounting weld nut	contamination on nut before welding	Spatter accumulated on weld nut during bracket welding	Weld nut not cover to avoid accumulation of weld spatter	No such provision of masking of weld nut during welding activity	Provide the facility
15	Front stab bar 'C' clamp position wrong	Front stab bar C clamp shifted from its actual position	Front stab bar C clamp tilted on resting fixture	Front Stab C clamp positioning fixture wear out	----	Correct the fixture
20	Chassis front stab bar mounting weld nut thread undersize	Stab bar mounting nut distort during welding process	Welding torque higher than specification	Welding torque not maintained in process	----	Welding to be optimized
21	Stab bar bolt tightening sequence wrong	Front stab bar bolt alignment with respect to Chassis weld nut & Stab bar C clamp gets disturb	After tightening C clamp one bolt alignment gets disturbed	Excess stress create at second bolt of C clamp	Stab bar bolt tightening sequence not appropriate	Change the sequence

6. Implement the corrective measures

Possible Remedies Identified

Legends

High ↑ Low ↓

Sr. No	Root Cause	Remedy Developed	Time to implement	Cost of Improvement	Easily reversible	Complexity	Decision
4	There is no mechanism to confirm all four nut torque	Provide the facility	Low	Low	Low	Low	Accepted
15	No such provision of masking of weld nut during welding activity	Provide the facility	Medium	Low	Low	Low	Accepted
16	Front Stab C clamp positioning fixture wear out	Correct the fixture	Medium	Low	Low	Low	Accepted
20	Welding torque not maintained in process	Welding to be optimized	Medium	High	Low	Low	Accepted
21	Stab bar bolt tightening sequence not appropriate	Change the sequence	Low	Low	Low	Low	Accepted

6. Implement the corrective measures

6.2 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Operator forget to tight front stab bar bolt/nut	Torque confirmation pokayoke provided	24.08.19	Sayali More	Closed.

Before

There is electric tool for Stab bar nut tightening, There is no torque confirmation of nut tightening, chances of tightening activity skipped- Marazzo model

Operator forget to tight front stab bar bolt/nut

After

Torque confirmation pokayoke provided on stage, Electric tool remains ON if all four nut not tightened, Torque confirmation display on stage

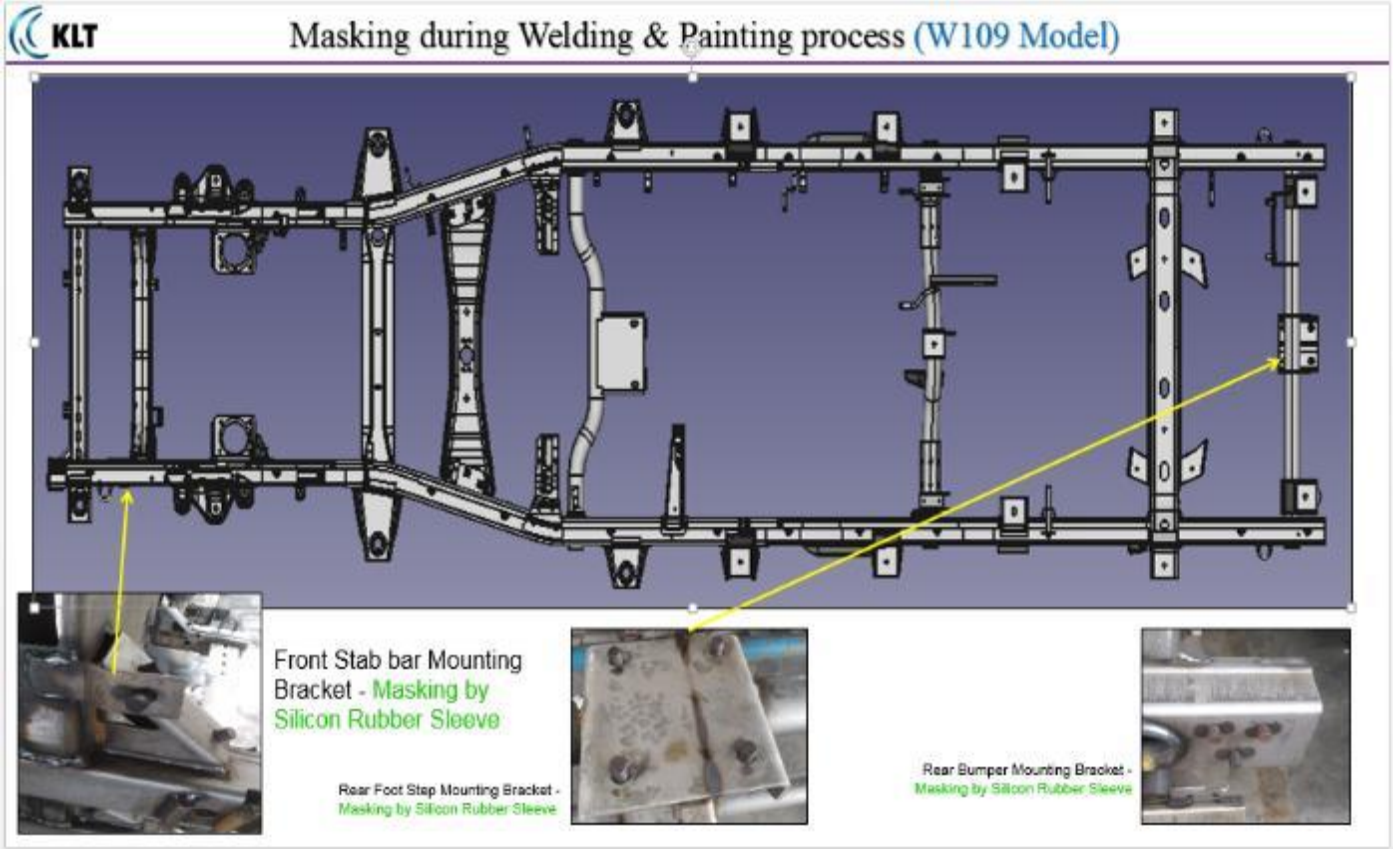
6. Implement the corrective measures

6.2 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Spatter in stab bar bolt mounting weld nut	Masking to be provide on nut to avoid spatter	05.08.19	Shailesh Kumar	Closed.

Before

Spatter in weld nut



6. Implement the corrective measures

6.2 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Front stab bar 'C' clamp position wrong	Front Stab bar C clamp positioning fixture to be correct	19.08.19	Santosh Jadhav	Closed.

Before

TPU Blocks for bush assembly fixture warn out & Crimp Tool Oversize by 1 mm

Front stab bar 'C' clamp shifted from its position

After

Metallic blocks provided to bush & C clamp assembly fixture & Crimp tool blocks replaced with new one

6. Implement the corrective measures

6.2 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Spatter in stab bar bolt mounting weld nut	Masking to be provide on nut to avoid spatter	10.08.19	Shailesh Kumar	Closed.

Before

Spatter in weld nut

After

Action plan: Bolt pre fitment hard MAGNA

Sr. No.	Probable cause	Action plan	Responsibility	Target date
1	TPG is not qualifying due to Contamination in nut threads before projection welding	1.Ultrasonic cleaning activity to be started at RTF 2. Masking process is started after projection welding which will help to remove contamination	1.RTF 2.Magna	1.10-Aug-19 2.19-Aug-19

6. Implement the corrective measures

6.2 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Difficult to pre fit front stab bar all four bolt	1. Welding parameter/ torque to be optimize	20.08.19	Shailesh Kumar	Closed.

Analysis: Bolt pre fitment hard

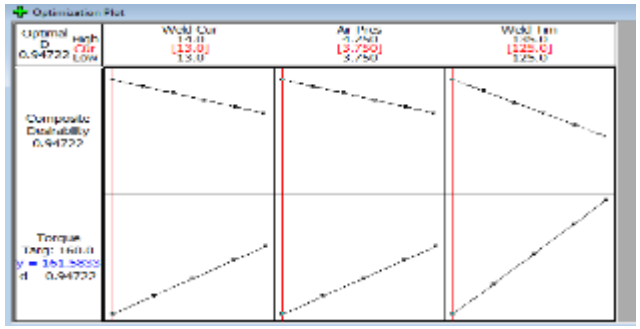
MAGNA

	Current (KA)	Weld Time (ms)	Pressure (Bar)	Squeeze Time (ms)	Hold Time (ms)	Specs: Torque Nm	TPG qualification
Range	21~22	250~270	5~5.1	150	220	Min 110 Nm	---
Actual	21.5	260	5	150	220	Greater than 250 Nm	30 Parts checked. In 4 parts TPG qualification is hard

Inference:
Torque is on higher side>250Nm against 110 Nm.
Parameter optimization to be done through DOE.

Before

After



Action plan: Bolt pre fitment hard

MAGNA

Sr. No.	Probable cause	Action plan	Responsibility	Target date
1	TPG is not qualifying due to Contamination in nut threads before projection welding	1.Ultrasonic cleaning activity to be started at RTF 2. Masking process is started after projection welding which will help to remove contamination	1.RTF 2.Magna	1.10-Aug-19 2.19-Aug-19
2	Welding parameters are on higher side of range leading to distortion in nut threads	Weld parameter optimization through DOE.	Magna	23-Aug-19

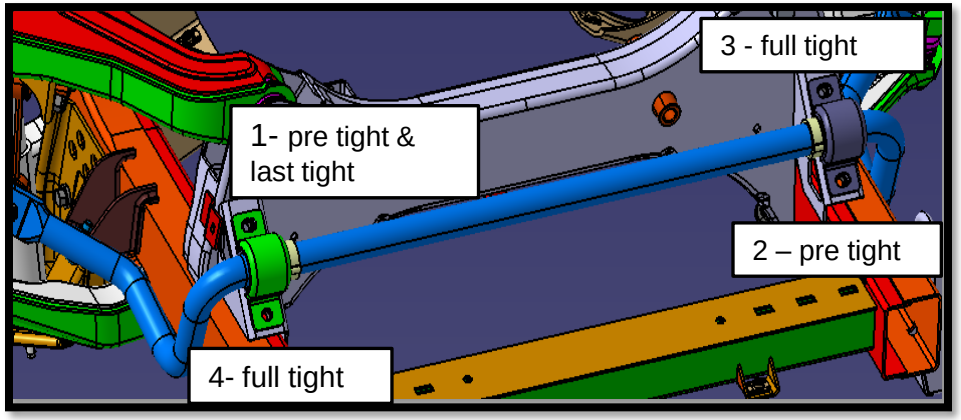
6. Implement the corrective measures

6.2 Actions Decided

Root Cause	Action Plan	Target Dt	Resp.	Status
Stab bar mounting bolt tightening sequence not OK	Stab Bar Bolt tightening sequence to be change	25.06.19	Ajit Chaudhari	Closed.

Before

After



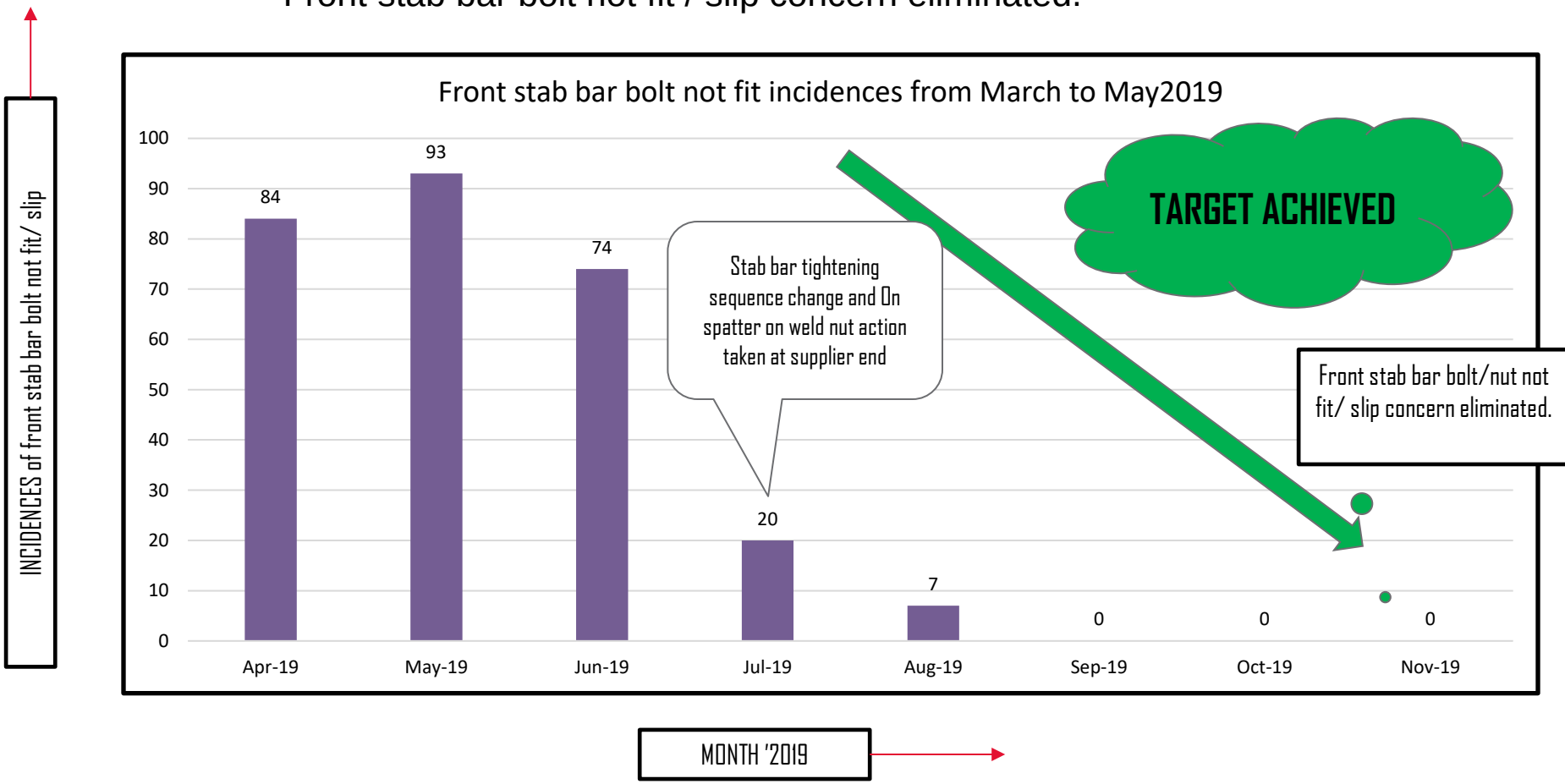
Stab Bar Bolt tightening sequence

Modified stab bar bolt tightening sequence

7. Check & confirm the results & effects

7.1 Verify Results:

Front stab bar bolt not fit / slip concern eliminated.



Inference- After taking PCA action, front stab bar bolt not fit/ slip concern eliminated.

7. Check & confirm the results & effects

7.3 Full Implementation Plan :

Action Plan	Target Dt	Resp.	Status
PFMEA - Front stab bar nut torque confirmation pokayoke to be provide	24.07.19	Sayali More	Closed
SOS - Stab Bar Bolt tightening sequence to be change	25.06.19	Ajit Chaudhari	Closed
Masking to be provide on nut to avoid spatter, Ultrasonic cleaning to be start to arrest contamination - SCAPA	05.08.19	Shailesh Kumar	Closed
Front Stab bar C clamp positioning fixture to be correct	19.08.19	Santosh Jadhav	Closed
Nut Welding parameter/ torque to be optimize	26.08.19	Shailesh Kumar	Closed
Process audit sheet - Sequence of tightening	03.08.19	Ajit Chaudhari	Closed
OPL - given to stage associate	03.08.19	Ajit Chaudhari	Closed
QA matrix	03.08.19	Ajit Chaudhari	Closed

Check & confirm the results & effects (Cont.....)

7.3 Tangible benefits:

- REWORK cost saving – Rs. 229238.7 Rs. Yr.
- Concern elimination from 3-4 cases/day to 0.
- DPT of 152 is reduced to zero.

7.3 Intangible benefits:

1. Operator fatigue reduced as stab bar bolt getting pre fit easily.
2. Customer satisfaction at next stage .

From: DHANOKAR RAJESH <DHANOKAR.RAJESH@mahindra.com>
Date: 24 December 2019 at 5:29:12 PM IST
To: CHAUDHARI AJIT <CHAUDHARI.AJIT@mahindra.com>
Subject: RE: cost estimation for MYB

Repair /Restoration Cost estimation		
Description	Units	
Rework time (Approx)	Min	45
Avg Concerns/Day	Nos	4.0
Total Required Rework time per day	Min	180
Effective working time	Min	465
Operator required /day	nos	0.39
Approx Operator cost/ day	Rs	1974
Operator cost /concern	Rs	764.13
Avg no. of working days	days	25
Operator cost /month	Rs	19103.2
Operator Cost/year	Rs	229238.7

8. Standardization

8.1 Updation of documents:

FMEA

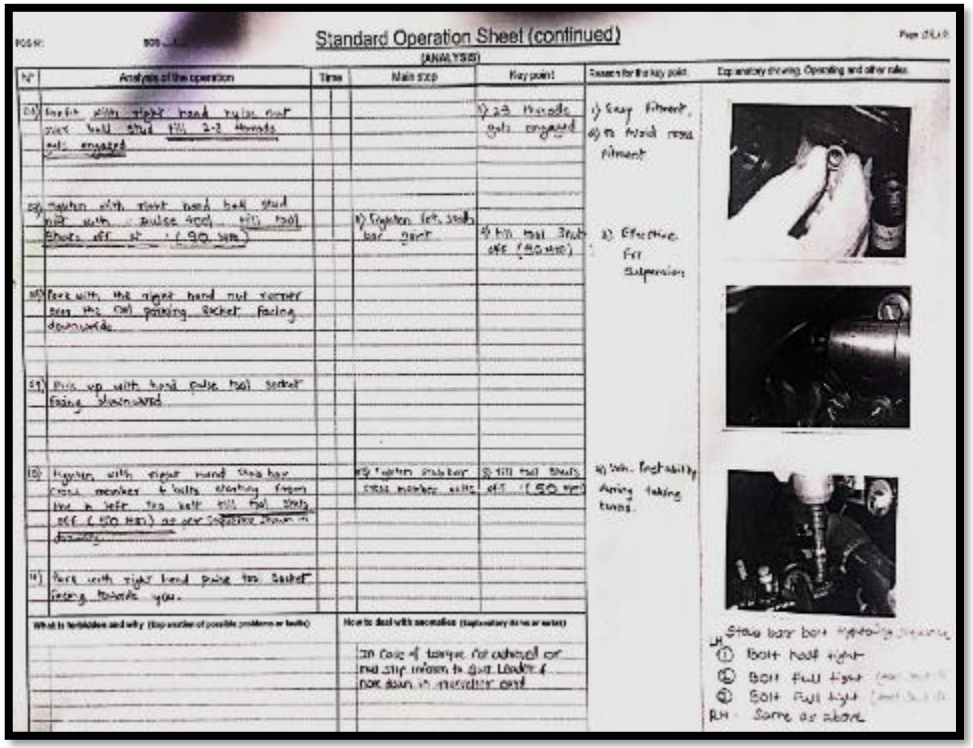
FMEA Updated for Stab bar nut tightening pokayoke

PU/ Shop : Scorpio / Xylo TCF			Optn Desptn: Front Stabiliser bar fitment on chassis			Reviewed by : Nikhil Hatedkar			Rev. No :	5	Doc No.: MQS/ODS/U321/TCF/04 01051					
Module/ Cell : TCF- Chassis			M/C Name & No.:			Approved by : Taj Shaikh			Rev. Date :	5.8.19	FMEA Date (Orig) :15.05.1					
Process Function	Requirements	Potential Failure Mode	Potential Effect(s) of Failures	Sev	Class	Potential Cause(s)/Mechanism(s) of Failures	Current Process Controls Prevention	Occur	Current Process Controls Detection	Detect	RPN	Recommended Action(s)	Responsibility & Target Completion Date	Action Results: Action Taken	Sev	Occ	Det	RPN
	1) Stab bar fitment on chassis bracket	not fit stab bar on chassis bracket	**Mfg & Assy :** 1) Further assembly is not possible 2) Rework at post station **End User :** Customer dissatisfaction due to rolling of vehical	5		Associate forget to install stab bar on chassis	Training and awareness given to stage Associate.	2	At IFS stage.	7	70							
	4) Pre fitment 'C' clamp position	1) C-clamp came out during installation	**Mfg & Assy :** 1) nut may get cross fitted. 2) nut thread may get damaged. 3) Rework at post station. **End User :** Stab bar not funtioning causing rolling of vehical.	6		Difficulty to prefit the nut because of less clearence.	Self certification. And awarness given to stage associate	2	Holes of 'c' clamp are not matching ,because of the less clearence to prefit the bolt.	7	84							
	6) Correct selection of nut	1) Wrong selection of stab bar nut	**Mfg & Assy :** leading to rework at post station. **End User :** Customer dissatisfaction due to rolling of vehical.	5		Associate select wrong stab bar.	Training and awareness given to stage Associate.	2	At IFS stage. And New IFS tightening stage	7	70							
	7) nut prefitment on c-clamp stud	1) nut cross/ slip	**Mfg & Assy :** eading to rework at post station. **End User :** Customer dissatisfaction due to rolling of vehical.	5		Associate did not prefit the nut	Training and awareness given to stage Associate.	2	Self check	7	70	Stab bar link nut tightning pokayoke started	AC/SM	Action closed	5	2	7	70

8. Standardization

8.1 Updation of documents:

SOS Updated :- For stab bar bolt tightening sequence


Standard Operation Sheet (continued)

(ANALYSIS)

N°	Analysis of the operation	Time	Main step	Key point	Reason for the key point. Explanatory drawing. Operating and other rules.
[illegible]	[illegible]			1) 2-3 threads gets engaged	1) Easy fitment. 2) To avoid cross fitment
[illegible]	Tighten with right hand ball stud nut with pulse tool till tool shuts off at (90 NM)		1) Tighten left stab bar joint	2) till tool shuts off (90 NM)	2) Effective for suspension
[illegible]	Fix with the right hand nut runner pass the nut holding socket facing downwards.				
[illegible]	Pick up with hand pulse tool socket facing downward				
[illegible]	Tighten with right hand stab bar cross member & bolts starting from the in left to right till tool shuts off (150 NM) as per sequence shown in drawing.		2) Tighten stab bar cross member with	3) till tool shuts off (150 NM)	3) Wh. fastability during taking turns.
[illegible]	Fix with right hand pulse tool socket facing towards you.				

What is forbidden and why (Explanation of possible problems or faults)

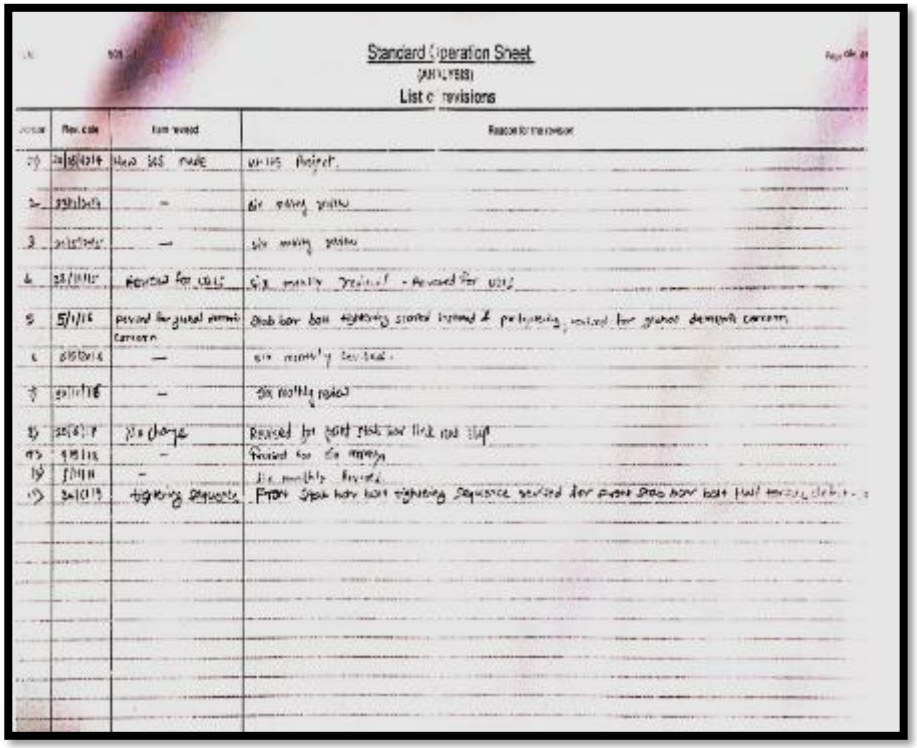
How to deal with anomalies (Explanatory drawing or notes)

In case of torque not achieved or nut slip inform to quality & not down in machine cycle

Stab bar bolt tightening sequence

LH
① Bolt hand tight
② Bolt Full tight
③ Bolt Full tight

RH - Same as above


Standard Operation Sheet

(ANALYSIS)

List of revisions

Issue	Rev. date	Item revised	Reason for revision
1)	28/8/2014	New SOS made	[illegible] Project.
2	[illegible]	–	[illegible]
3	[illegible]	–	[illegible]
4	23/[illegible]	Revised for [illegible]	Six monthly [illegible] - Revised for [illegible]
5	5/1/16	Revised for [illegible]	Stab bar bolt tightening started inside & [illegible] revised for [illegible]
6	[illegible]	–	Six monthly [illegible]
7	[illegible]	–	Six monthly [illegible]
8)	[illegible]	[illegible] change	Revised for [illegible]
9)	[illegible]	–	Revised for [illegible]
10)	[illegible]	–	Six monthly [illegible]
11)	[illegible]	tightening sequence	From Stab bar bolt tightening sequence revised for [illegible]

Work station SOS

8. Standardization

8.1 Updation of documents:

SUPPLIER'S CORRECTIVE ACTION PLAN- FRONT STAB BAR

TO BE FILLED BY M&M

M & M Ltd. Automotive Sector	**SUPPLIER'S CORRECTIVE ACTION PLAN (SCAP)**	PU Name : Mr.Shailesh Kumar Sr.No: 'CCA-14-19 Date: 9/20/2019

Supplier Allevard-iAi Suspensions Pvt. Ltd. — **Supplier Code:** DA221X
Part Description: Stabiliser Bar Front — **Part No.:** 0401EBB00371N
When Observed (Date / Period) Assly Line — **GR/Invoice No.:** NA

A PROBLEM STATEMENT Should Answer — **ISSUE REPORTED :** REPEAT ☐ / 1'ST TIME ☑

What (is the problem? (Expression by customer)	
Who is affected	M & M - Assly Line
Where Location	M & M Nashik Assly Line
When Since	20/09/2019
Why is the problem	Bracket position shifted
How much Qty	01 No

Bracket position shifted by 1 mm

(Problem statement should answer 5W 1H ie What, When,Who,Where, & How.
(Attach sketch / data for details where required)

TO BE FILLED BY SUPPLIER

B REASON FOR SENDING NOT OK PARTS : (Route cause for non detection at vendor end)

5 WHY ANALYSIS (ANALYSIS FOR NOT DETECTING DEFECT AT SUPPLIER END)

WHY	WHY	WHY	WHY	WHY
Bracket position shifted	Crimp CD oversize by 1mm observed after layout inspection	**Crimp Tool CD Oversize by 1 mm**	Bracket / Bush Assembly fixture - Crimp Locating blocks worn out	**Bracket / Bush Assembly fixture - Block made by TPU material**

C ACTION TAKEN TO DISPOSE NON -CONFORMING PARTS:

Reworked at Assly line

E ROOT CAUSE ANALYSIS OF PROBLEM: ☑ YES ☐ NO

Machine
Is the specified M/C used ? ☑ YES ☐ NO
Is M/C in good condition ? ☐ YES ☑ NO
Is it a substitute M/C ? ☐ YES ☑ NO
Are M/C parameters known ? ☑ YES ☐ NO
Parameters are respected ? ☑ YES ☐ NO
***

Man
Is operator trained ? ☑ YES ☐ NO
Is operator is under training ? ☐ YES ☑ NO
Is operator temporary ? ☑ YES ☐ NO
Is it due to Bad Rework ? ☐ YES ☑ NO
***

Measurement
Are instruments / gauges calibrated ? ☑ YES ☐ NO
Are Control Methods Capable to detect Non conformity? ☑ YES ☐ NO
Are specified Control Methods used ☐ YES ☑ NO
***

Bracket position shifted by 1 mm

Method
Are Control Plan/SOP available near M/C? ☐ YES ☑ NO
Are Control Plan respected ? ☑ YES ☐ NO
Is the defect part was detected by operator / inspector & later mixed with production ? ☐ YES ☑ NO
***

Material / Tooling
Is the specified Raw Matl./Tooling used ? ☑ YES ☐ NO
Is the Raw Matl./Tooling Supplier change ? ☐ YES ☑ NO
Is the heat treatment method change ? ☐ YES ☑ NO
***
1.Crimp Tool CD Oversize by 1 mm.
2.Bracket / Bush Assembly fixture - Block made by TPU material

Environment
Is process needs environmental control? ☐ YES ☑ NO
Is it implemented ? ☐ YES ☐ NO
Does it conform ? ☐ YES ☐ NO
***
Not Applicable

*** Space provided for any other cause

Analysis Conclusion:-

Crimp Tool Oversize by 1 mm — 20%
Bush assembly fixture - TPU Block worn out — 80%

F CORRECTIVE ACTION PLAN: (Actions planned to avoid recurrence)

Sr. No.	Major causes identified	Action Plan	Action By	Target Date	Comp. On
1	**Bush assembly fixture - TPU Block worn out**	Metallic locating block provided to avoid TPU block	Santosh Choudhary	9/20/2019	9/30/2019
2	**Crimp Tool Oversize by 1 mm**	Crimp tool blocks replaced with new one	Santosh Choudhary	9/20/2019	9/30/2019

G ACTION TAKEN TO PREVENT RECURRENCE

1	Tool History card - checkpoint added to check worn out metallic block	Santosh Choudhary	9/20/2019	9/30/2019

H CORRECTIVE ACTION PLANNED IN SIMILAR PARTS/CASES ACROSS DEPT./ PRODUCTS/ PLANT (HORIZONTAL DEPLOYMENT)

1	P601 Jeeto	Santosh Choudhary	9/20/2019	9/30/2019
2	P607 Jeeto	Santosh Choudhary	9/20/2019	9/30/2019

I Following quality documents changed due to implemented action plan :
(mention doc. no.)

Process Flow Chart	☐	**Gauge Drawing**	☐
PFMEA	☐	**Fixture Drawing**	☐
Control Plan	☐		
Work instruction / SOP	☐		

L EFFECTIVENESS MONITORING : - To be filled by M&M Representative (after 3 months)

Are actions taken are sufficient to fix the problem	☐ YES	☐ NO
Are actions planed completed / implemented	☐ YES	☐ NO
Is problem repeated in lots received after SCAP issue Date	☐ YES	☐ NO

FOR MAHINDRA & MAHINDRA (A.S.)	**SUPPLIER'S REPRESENTATIVE**
Signature / Date	**Signature / Date** Pravin Suryawanshi /30.09.2019

MQS/SCAP/ F/01 **Rev.02** **6/25/2007**

8. Standardization

8.1 Updation of documents:

SUPPLIER'S CORRECTIVE ACTION PLAN - CHASSIS

TO BE FILLED BY M&M

M & M Ltd.
Automotive Sector

SUPPLIER'S CORRECTIVE ACTION PLAN (SCAP)

PU Name : Scorpio
Sr.No:
Date: 24.08.2019

Supplier:- KLT AUTOMOBILE & TUBULAR PRODUCTS LTD — **Supplier Code:** DK055
Part Description: Scorpio SC/DC W109 — **Part No.:0201AM00850N**
When Observed (Date / Period) : 22.08.2019 — **GR/Invoice No.:**

A PROBLEM STATEMENT Should Answer — **ISSUE REPORTED : REPEAT ☐ / 1'ST TIME ☑**

What (is the problem? (Expression by customer))
Who is affected.
Where location
When Since
Why is the problem.
How much Qty

Problem Statement:
Stab bar mounting bolt slip
M&M Nashik
22.08.2019
01
Nut offset

(Problem statement should answer 5W 1H ie What, When,Who,Where, & How.
(Attach sketch / data for details where required)

TO BE FILLED BY SUPPLIER

B REASON FOR SENDING NOT OK PARTS : (Route cause for non detection at vendor end)

5 WHY ANALYSIS (ANALYSIS FOR NOT DETECTING DEFECT AT SUPPLIER END)

WHY	WHY	WHY	WHY	WHY
Stab bar bolt slip	Burr and spatter inside nut threading	Nut threading not checked at incoming inspection	Masking of bracket not done	

C ACTION TAKEN TO DISPOSE NON -CONFORMING PARTS:

1) Defective chassis reworked and verified at customer end
2) All stock chassis checked & verified.(In plant)

E ROOT CAUSE ANALYSIS OF PROBLEM:

Machine
Is the specified M/C used ? ☑ YES ☐ NO
Is M/C in good condition ? ☑ YES ☐ NO
Is it a substitute M/C ? ☐ YES ☑ NO
Are M/C parameters known ? ☑ YES ☐ NO
Parameters are respected ? ☑ YES ☐ NO
***

Man
☑ YES ☐ NO
Is operator trained ? ☑ YES ☐ NO
Is operator is under training ? ☐ YES ☑ NO
Is operator temporary ? ☐ YES ☑ NO
Is it due to Bad Rework ? ☐ YES ☑ NO
***

Measurement
Are instruments / gauges calibrated ? ☑ YES ☐ NO
Are Control Methods Capable to detect Non conformity? ☑ YES ☐ NO
Are specified Control Methods used ☑ YES ☐ NO
***

Method
Are Control Plan/SOP available near M/C? ☑ YES ☐ NO
Are Control Plan respected ? ☑ YES ☐ NO
Is the defect part was detected by operator / inspector & later mixed with production ? ☐ YES ☑ NO
***

Material / Tooling
Is the specified Raw Matl./Tooling used ? ☑ YES ☐ NO
Is the Raw Matl./Tooling Supplier change ? ☐ YES ☑ NO
Is the heat treatment method change ? ☐ YES ☐ NO
***

Environment
Is process needs environmental control? ☐ YES ☑ NO
Is it implemented ? ☐ YES ☑ NO
Does it conform ? ☐ YES ☑ NO
***

Analysis conclusion:-
We suspected that in defective chassis bolt slip due to burr or spatter inside thread

F CORRECTIVE ACTION PLAN: (Actions planned to avoid recurrence)

Sr. No.	Major causes identified	Action Plan	Action By	Target Date	Comp. On
1	In defective chassis bolt slip due to burr or spatter inside thread	1.Sampling inspection increased from 10 nos. to 50 nos. in a lot (Random) 2.Corrective action taken from tier II supplier	Swapnil S.	24.08.2019	24.08.2019
2	Nut tapping not checked / done at tapping stage	Training and awareness given to operator	Amol S.	24.08.2019	24.08.2019

G ACTION TAKEN TO PREVENT RECURRENCE

1.Sampling inspection increased from 10 nos. to 50 nos. in a lot (Random)
2.Corrective action taken from tier II supplier
3.Training and awareness given to operator

H CORRECTIVE ACTION PLANNED IN SIMILAR PARTS/CASES ACROSS DEPT./ PRODUCTS/ PLANT

I Following quality documents changed due to implemented action plan :

(mention doc. no.)
Process Flow Chart ☐
PFMEA ☐
Control Plan
Work instruction / SOP ☑
Gauge Drawing ☐
Fixture Drawing ☐

L EFFECTIVENESS MONITORING : - To be filled by M&M Representative (after 3 months)

Are actions taken are sufficient to fix the problem ☐ YES ☐ NO
Are actions planed completed / implemented ☐ YES ☐ NO
Is problem repeated in lots received after SCAP issue Date ☐ YES ☐ NO

FOR MAHINDRA & MAHINDRA (A.S.)
Signature / Date

SUPPLIER'S REPRESENTATIVE
Yogeshwar vengurlekar
Signature / Date

MQS/SCAP/ F/01 — Rev.02

KLT

Concern	ICA	RC	PCA
Threading Issue- Bolt per fitment not possible, Spatter on nut	1. Dedicated Inspector deployed to check nut threading bracket wise. 2. After painting third party senior inspector deployed till concern resolve.	1. Weld distortion during welding on nut. 2. Weld spatter on threading area during weld process.	1. Project started for nut masking during welding process 2. For threading Route card started. 3. Nut bracket wise number of days without defect monitoring started. 4. For Projection welding die standardization bracket hole size standardization in process. 5. Hourly masking Process adherence audit by QA Engineer started.

8. Standardization

8.1 Updation of documents:

SUPPLIER'S CHECK SHEET

KLT

Pre delivery inspection report

8. Standardization

8.1 Updation of documents:

SUPPLIER'S CHECK SHEET

KLT

Masking Audit Check sheet at Amey

KLT Automotive & Tubular Products Ltd

Masking Audit Check Sheet

Sr. No.	Nut Position	Nut Size	Qty	Date				
				09/08/19				
1	Condencern mtg Brkt	M8	2	✓				
2	Apron Mtg Bracket	M8	2	✓				
3	Fuel clip bracket	M6	1	×				
4	Steering Mtg. Brkt	M12	2	✓				
5	Brake Tee Bracket	M6	1	×				
6	Engine Mounting Bracket	M8	8	✓				
7	Front Stabilizer bracket	M10	4	✓				
8	Fuel Tank Front	M8	2	✓				
9	Fuel tank Rear Inner	M8	2	✓				
10	Fuel tank Rear Outer	M8	2	✓				
11	Rear Pivot Bracket	M8	2	✓				
12	Rear Pintal Hook Bracket	M10	4	✓				
13	Spare Wheel Winch brkt	M6	2	×				
14	Rear Bumper Bracket	M8	6	✓				
			40					
Checked By								

Varify By

Format No. FF/QM520/08 Rev-00

8. Standardization

8.1 Updation of documents:

SUPPLIER'S CHECK SHEET- WELD PARAMETER

Confirmation run & Results after welding parameter Optimization

MAGNA

Sl.No	Current (KA)	Weld Time(msec)	Squeeze Time(m sec)	Pressure (Bar)	Hold Time (m sec)	Torque (Nm)	Inspection with TPG
1	13.00	125	150	3.75	220	180	Found Ok
2	13.00	125	150	3.75	220	160	Found Ok
3	13.00	125	150	3.75	220	160	Found Ok
4	13.00	125	150	3.75	220	180	Found Ok
5	13.00	125	150	3.75	220	180	Found Ok
6	13.00	125	150	3.75	220	170	Found Ok
							Found Ok
							Found Ok
							Found Ok
							Found Ok
							Found Ok
							Found Ok
							Found Ok
							Found Ok
							Found Ok
							Found Ok
							Found Ok
							Found Ok
							Found Ok
							Found Ok
							Found Ok
							Found Ok
							Found Ok
							Found Ok
							Found Ok
25	13.00	125	150	3.75	220	170	Found Ok
26	13.00	125	150	3.75	220	170	Found Ok
27	13.00	125	150	3.75	220	170	Found Ok
28	13.00	125	150	3.75	220	170	Found Ok
29	13.00	125	150	3.75	220	165	Found Ok
30	13.00	125	150	3.75	220	175	Found Ok

Conclusion: Optimized parameter details & results

MAGNA

	Current (KA)	Weld Time (ms)	Pressure (Bar)	Squeeze Time (ms)	Hold Time (ms)	Specs: Torque Nm
Range	13~14	125~135	3.75~4.25	145~155	215~225	Min 110 Nm
Optimized parameters	13	125	3.75	150	220	Actual: 160~190 Nm

11/10/2019 Author: Add Name Disclosure or duplication without consent is prohibited 15

8. Standardization

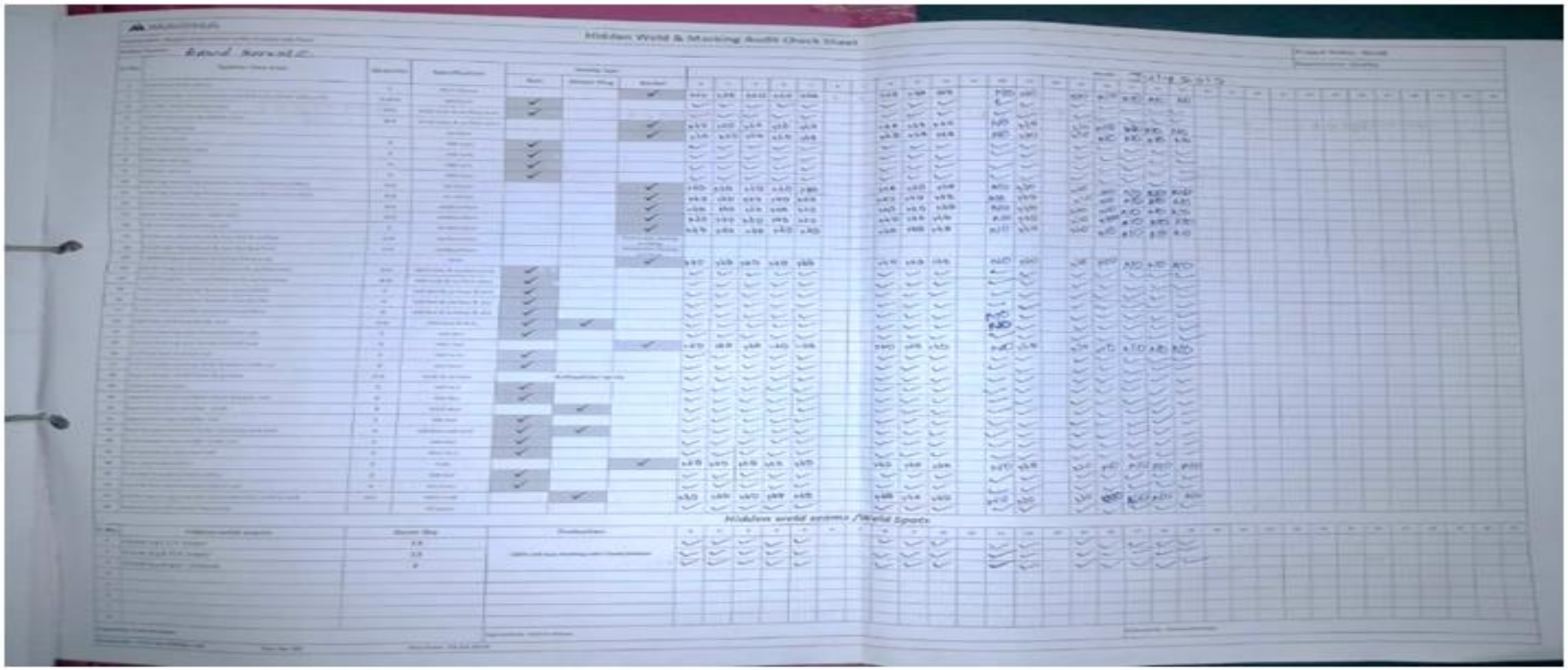
8.1 Updation of documents:

SUPPLIER'S CHECK SHEET- MASKING TO AVOID SPATTER

Action Plan for spatter reduction

MAGNA

Masking Audit Check sheet



8. Standardization

8.1 Updation of documents:

Process audit sheet modified for Stab bar bolt tightening sequence

PROCESS AUDIT SHEET BWT 07

BWT No.	Stage / Fitment process	What to check	Model	DATE						
7	Traveller card filling	Trav card not filled before chassis drop on line - check cards in storage	All							
7	Steering rack fitment	a) Protecting caps available on rubber boot b) Bolt pre fitment & tightening sequence asper SOS	All							
7	Lower link fitment	a) W105 - White sticker lower link b) Xylo/Quanto/103 - Yellow sticker (Cayman) on lower link.	All							
7	Bump stop fitment	Properly inserted & match marking	W105							
7	Hand brake cable fitment	a) Routing above upper link. B) Hand brake cable clip Lh & RH correctly used	All							
7	GR tool tigthening	Don't apply the compression fixture and see weather the GR tool starts or not	All							
7	Axle selection on line	Operator checking axle part no. wrt build sheet.	All							
7	Rear suspension tightening	Compression fixture usage & not to be removed before complete torquing.	All							
7	Coil spring fitment	a)Selection as per colour identification - check operator awareness. Sticker clor -Xylo - Yellow,W103 - Green, Quanto - Pink, W105 - White. C) Coil spring's end to be put on marking provided on pad.	All							
7	Coil spring pad	a) Selection as per model. B) Coil springs end to be put on marking as per model	All							
7	Rear axle TVD tuner	a) Correct orientation of prop shaft written side towards propeller b) Screw to be prefit 2-3 threads before tightening.	All							
7	Power str pipe fitment	Protective caps present, tightening sequence as per SOS	Xylo & quanto							
7	Shock ab.fitment	a) Selection as per colour identification b) Marking after bolt tightening c) Brake back application near engine drop stage - marking on shock absorber	All							
7	Z-Stage monitoring	How document's displayed on the board, Doc must be updated with reference to RFI,NOVA-C,buy off,SCR concerns, Awarness of stage opearators & Operator as per Pool matrix, callibration of tools used at Z stage.	ALL							
7	General points	JH / 5S round points :- 1)No mix up of material in bin 2)Tool park at tool parking not in bin 3) Hand gloves not put in material bins. 4)Contamination for oil/Diesel/Coolant 5) Layout marking but tool/material not in place	All							
7	Stab. Bar fitment	a) Protecting caps available on rubber boot. b) Selection as per model c) Ensure bolt tightening sequence d) Ensure use of special tool for stab bar bolt pre fitment	All							
7	Fuel tank fitment	1) Check all bolts pre fitment and tightening sequence as per SOS 2) No loose fitment of fuel tank mounting bkt 3) Marking on bolts	All							
7	spare wheel fitment	Writing make initials on build sheet.	All							
7	LCA fitment	Compression fixture is getting remove only after complete fitment	All							
7	LCA Compression	Compression distance checking by Go- No Go gauge once in shift	All							
7	IFS fitment	FT shock absorber fitment compression fixture to be used. Use of pokayoke. Use of IFS degree setting fixture For W105	All							
7	Z-Stage monitoring	How document's displayed on the board,Doc must be updated with reference to RFI,NOVA-C,buy off,SCR concerns,Awarness of stage opearators,callibration of tools used etc.	ALL							

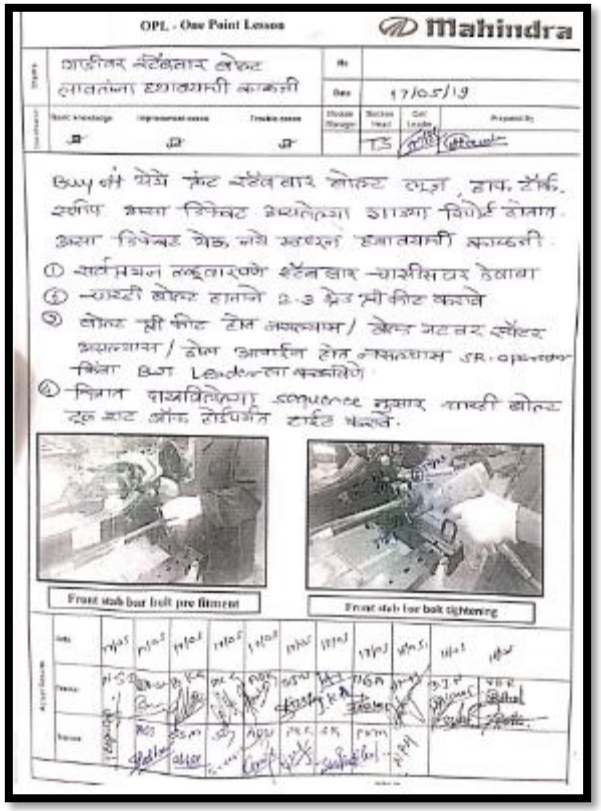
Front stab bar bolt tightening sequence checking & ensuring use of special tool for stab bar bolt pre fitment added

8. Standardisation

8.2 Training:

Action Plan	Target Dt	Resp.	Status
OPL and Training given to all Stage associate for use of special tool to pre fit stab bar bolt & to follow stab bar tightening sequence	01.07.19	Ajit Chaudhari	Closed

On Job training given to all stage associate for new tightening process



Training OPL

8. Standardization

8.1 Updation of documents:

5. QA Matrix – **Front stab bar bolt not fit/ Slip concern**

QA MATRIX

PU NAME	STCF	INITIALIZATION DATE						Global Quality Guarantee level.	
DEPT NAME		SIGNATURES							
SECTION NAME	CHASSIS	Dept Head	Rahul Narkar					Section	Chassis
		Module Manager	Taj Shaikh					Plant	Nasik
		Section Manager	1 SS	2 AC	3 MP	4 SB		Revision date	10.9.19

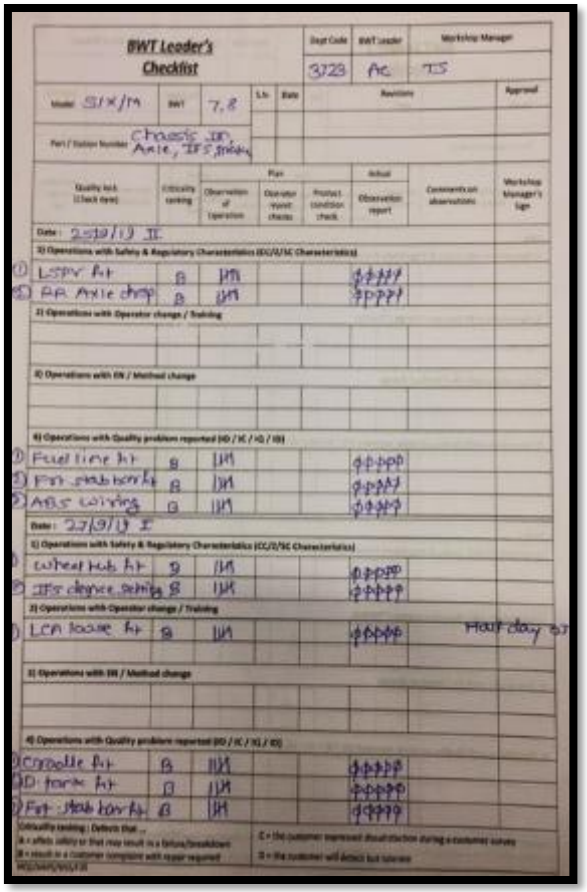
Workstation No.	Process description (POS)	Date	Failure modes	Severity	Prev. occurence	Potential decfect	Inspection process – Section: Chassis IN	Rear axle fitment	LCA fitment	Front suspension fitment	Brake	Propeller shaft fitment	I shaft fitment	Engine Drop	Exhaust fitment	Centre bering fitment	Chassis punching	Interim buyoff	Oil filling stage	Tyre and OIL	Buyoff	Outside section: Chassis drop stage	Body drop stage	Console stage	Filling stage	FAI Buyoff	CAI Roller	CAI underpit	Minor mechanical	Major mechanical	Wrap	Guarantee value: Workstation	Section	Outside section	PLANT	Faults recorded: Section	Outside section	CUSTOMER	Quality guarantee level: WORKSTATION	Section	Outside section	PLANT	Comments
																																					1M	1M	3M					
1	Front stab bar	10.9.19	Bolt half torue/ Bolt missing/ Bolt loose	V 1 5						⊙ 3								□ 1		⊙ 3						□ 1		□ 1				3	7	2	12	3	0	0	NOK	OK	NOK	OK	

8. Standardisation

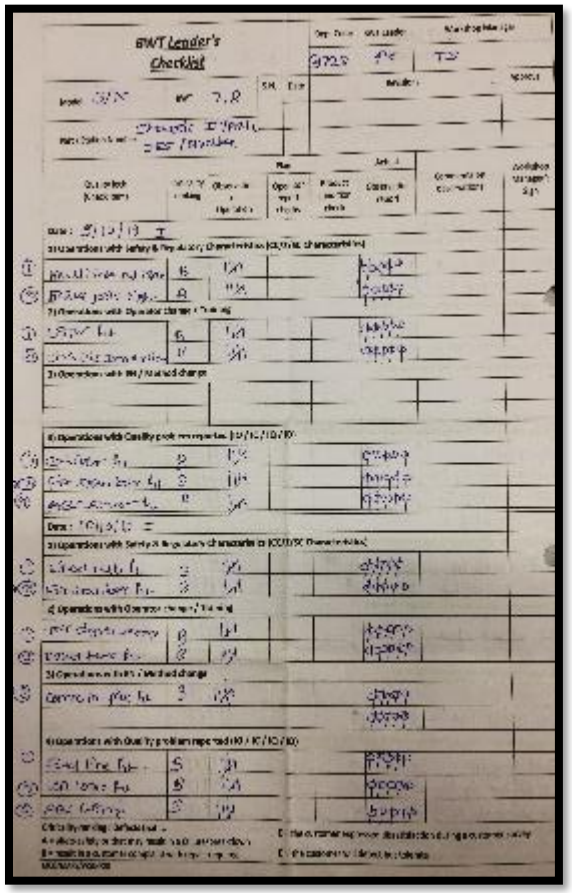
8.3 Additional Audit / Check

Point taken in daily supervisor check sheet –till NOV'19

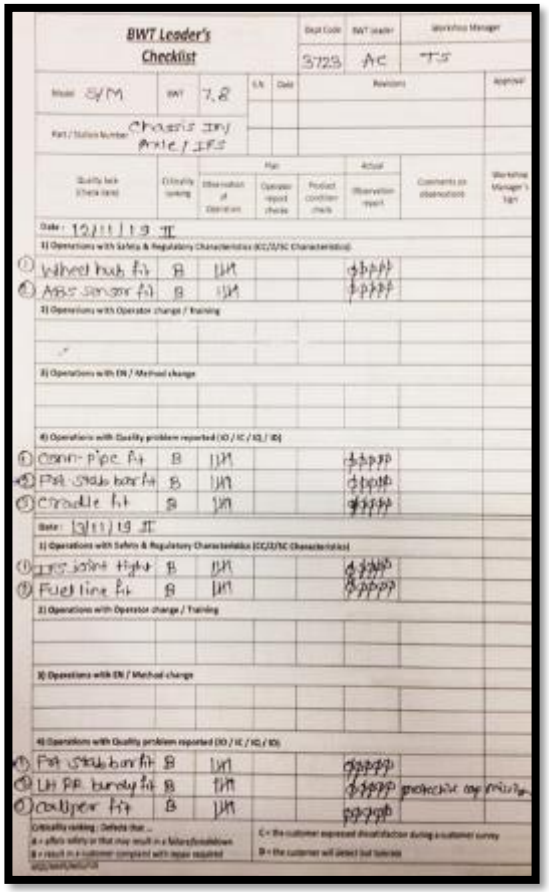
SEP



OCT



NOV



9. Summarise & Plan future actions

9.1 Conclusion:

Front stab bar bolt not fit/ Slip concern in all model resolved by following actions.–

1. Stab bar bolt tightening sequence changed
2. Masking & ultrasonic cleaning activity started to avoid spatter in weld nut
3. Chassis stab bar mounting weld nut welding torque/ parameter corrected
4. Front stab bar 'C' clamp positioning fixture corrected
5. Poka-yoke started for Marazzo stab bar nut tightening

9.3 Horizontal Deployment:

Masking activity to avoid spatter for all Chassis weld nut

9.3 Lessons Learnt:

THINGS GONE WRONG: Masking on nuts during painting which is necessary thing but was skipped by Supplier.

THINGS GONE RIGHT: Putting Pokayoke for confirmation of all the nuts are torqued helped to deskill the operator.

Thank You